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RESEARCH

Simulation-based qualification of repair and remanufacturing operations through DED technology of Ti6Al4V parts

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Abstract

Qualification of repair and remanufacturing operations through directed energy deposition (DED) offers a sustainable pathway to restore and even enhance the performance of high-value metallic components. By employing additive manufacturing techniques to deposit material onto damaged parts, this approach minimizes downtime and extends the lifecycle of critical components such as molds, dies and tooling. However, integrating newly deposited material with the original substrate poses challenges related to residual stress formation, distortion, and alterations of the microstructural. In this study, qualification is addressed from two interrelated perspectives: (1) the qualification of the component-ensuring dimensional accuracy, minimal residual stresses and preserved metallurgical integrity; and (2) the qualification of the DED process-focusing on buildability through optimized power supply, scanning speed and dwell times. Because extensive experimental testing on high-value parts is impractical and often destructive, simulation studies are the only viable means to qualify repair operations. A simulation-assisted dynamic power control strategy is developed to maintain consistent melt-pool behavior through feedback-based power modulation. The model is calibrated and validated using finite element simulations and targeted experiments. Results demonstrate that power modulation and inter-layer dwell time effectively stabilize the melt-pool and reduce heat accumulation, improving dimensional accuracy, microstructural uniformity, and hardness distribution. However, these benefits are accompanied by increased inter-layer residual stresses due to enhanced thermal gradients. The study therefore establishes the trade-offs between thermal control, mechanical response, and microstructural control in DED-based remanufacturing, providing practical insight into selecting appropriate process parameters for qualification.

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Abbreviations

FE	Finite element
FEs	Finite elements
AM	Additive manufacturing
SM	Subtractive manufacturing
MAM	Metal additive manufacturing
AMR	Adaptive mesh refinement and coarsening
DED	Direct energy deposition
TTT	Temperature-time-transformation
JMAK	Johnson–Mehl–Avrami–Kolmogorov
GCode	RS-274
ML	Machine learning
HTC	Heat transfer coefficient

1 Introduction

Qualification of repair and remanufacturing operations can be divided into two distinct complementary branches. First, qualification of the component ensures that the repaired or remanufactured high-value part meets strict criteria of dimensional accuracy (minimizing warpage), residual stresses (keeping them as low as possible) and metallurgical integrity. Second, qualification of the Direct Energy Deposition (DED) process concentrates on buildability aspects-optimizing parameters such as power, speed and dwell times to guarantee process stability and repeatability. The present work addresses the relationship between both aspects, i.e. whether the conditions that optimize the DED process also yield a qualified component, since process efficiency does not necessarily translate into optimal mechanical or metallurgical performance.

In the context of the circular economy, remanufacturing has emerged as a sustainable approach to extend the lifecycle of high-value metallic components by restoring them to performance levels comparable to or even exceeding those of new parts. Remanufacturing by DED combines Additive Manufacturing (AM) deposition with conventional reconditioning methods to repair or upgrade end-of-life molds, dies, and turbine blades. This approach reduces lead times, retooling, and waste generation while improving repair precision and material efficiency. Figure 1 illustrates a remanufactured compressor before and after post-processing using advanced DED technology.

Recent research has further expanded the role of AM within remanufacturing workflows by integrating AM and Subtractive Manufacturing (SM) processes into hybrid frameworks. As shown in [1], combining AM with machining enables the regeneration of end-of-life metallic parts, transforming scrap components into added-value products. The work presented in [2] explores process planning strategies for such hybrid manufacturing systems, highlighting the need to properly sequence additive and subtractive operations to ensure dimensional restoration and surface quality. From a sustainability perspective, [3] reports that AM-SM combination for remanufacturing significantly reduces material waste and energy consumption compared to conventional repair methods. The present study complements these efforts by addressing the numerical qualification of repair operations through simulation-based control of process parameters, establishing the relationship between buildability and component performance.

While hybrid AM-SM frameworks have advanced the process integration of remanufacturing, the qualification of the deposited material and substrate remains a key challenge. In conventional AM applications, the primary focus is on achieving high-quality parts with enhanced mechanical performance, dimensional accuracy, surface finish and material integrity. However, in the remanufacturing context, it is equally critical to ensure that both the newly deposited material and the pre-existing substrate maintain optimal performance. The quality of the interface between the repair material and the substrate is paramount, as any issues in bonding strength or thermal compatibility can compromise the overall integrity of the component. Residual stress concentrations are a common challenge at the substrate-repair interface, as the solidification of the deposited layer exerts compressive forces on the underlying material, inducing complex stress states [4]. Simplified models such as the three-bar analogy proposed in [5] highlight the influence of temperature gradients and the Yield Temperature (YT), i.e. the temperature at which plasticity initiates, showing that



(a) (b)

Fig. 1 Remanufactured compressor using DED technology (Godiz Servicios Industriales). **a** Before the post-processing. **b** After the post-processing

reducing substrate restraint can effectively mitigate residual stress formation.

Efforts to manage these stresses resulted in substrate design modifications. Numerical studies [6] show that reducing substrate stiffness or adding geometrical features such as edge rounding can decrease local stress concentrations. However, less stiff substrates may induce higher distortion, introducing a trade-off between stress relief and dimensional stability. The smart-substrate design proposed in [7] reduces local stiffness by adding controlled grooves, functioning similarly to expansion joints. While this configuration effectively relieves stress, it may compromise the structural integrity of the substrate, i.e. the repaired/remanufactured component.

Beyond substrate design, process parameters and thermal management techniques are crucial in controlling residual stresses. Studies such as [8–11] emphasize the influence of scanning sequence, vector length and power strategy on the distribution and magnitude of residual stresses. Complementary approaches-such as pre-heating the substrate or applying post-heating treatments in the range of $600 - 700\,^{\circ}\mathrm{C}$ [12, 13]-have been shown to drastically reduce tensile stresses, offering a practical means of stress management during and after the repair process. Moreover, the mitigation of distortions, a related consequence of residual stresses, benefits from both thermal treatments and optimized process parameters, as reported in [14, 15].

Dwell time is a widely adopted strategy in additive manufacturing to prevent overheating and to allow sufficient cooling between layers to improve buildability. In reference [16], the authors demonstrate that while adding dwell time generally helps manage the thermal history of a build, its effects can vary significantly with the material. For instance, in Inconel®625 builds, incorporating longer dwell times (20–40 s) effectively reduces both the accumulated distortion and residual stress by promoting a more uniform cooling process. In contrast, for Ti–6Al–4V, which is prone to higher peak temperatures due to its lower thermal conductivity and phase transformation behavior, eliminating dwell time leads to an 81.7% reduction in early-layer distortion and lower overall residual stress, despite the associated higher peak temperatures.

The influence of process parameters extends beyond thermal stress management to microstructural evolution. The thermal cycles dictating stress accumulation also govern solidification and phase transformations. Melt-pool temperature control can enhance microhardness and reduce variability in dendrite spacing [17], while cooling rate directly affects α-variant distribution [18]. Faster cooling promotes weaker texture and finer α colonies, whereas slower cooling favors coarser Widmanstätten structures. Maintaining consistent melt-pool conditions is thus essential for predictable mechanical properties [17, 19]. Power modulation strategies-either simulation-assisted [20] or closed-loop [21–23]-have been developed to stabilize melt-pool dynamics and improve build uniformity.

Recent developments in Machine Learning(ML) have further advanced predictive capabilities for AM. Applications include thermal history estimation [24, 25], parameter optimization [26, 27], defect prediction [28, 29], and part-quality assessment [30, 31]. Machine-learning models have also been applied to predict distortions [32, 33] and material properties [34, 35]. However, these models require extensive datasets for training and validation. High-fidelity simulations [36–40] offer a scalable alternative to experimental data, enabling the generation of comprehensive datasets for predicting thermal, mechanical, and microstructural responses.

Similar simulation-driven frameworks have been successfully implemented in subtractive manufacturing processes, particularly in machining, where simulation-driven optimization is used to balance efficiency and tool integrity. For example, [41] presents a finite-element-based multi-objective optimization of machining parameters, integrating thermal-mechanical simulations with evolutionary algorithms to minimize tool wear and maximize material removal rate. Such studies demonstrate the potential of simulation-assisted control and optimization, providing a methodology in line with the present work, which extends these concepts from subtractive to additive procedures-applying them to the qualification of repair and remanufacturing operations via DED.

Despite these advances, the qualification of repair and remanufacturing operations remains challenging due to the cost and risk associated with experimental validation on high-value parts. Intrusive testing may alter or damage components through residual stress induction, warpage, or microstructural modification. Simulation-based methods thus provide a non-destructive alternative for predicting performance outcomes under various process conditions, ensuring reliability and repeatability while preserving component integrity.

Motivated by these challenges, this work develops a simulation-based framework for the qualification of repair and remanufacturing operations via DED of Ti-6Al-4V. The methodology integrates experimental calibration and process optimization to achieve a predictive control of melt-pool behavior. The calibration, performed using the `DAKOTA` optimization tool [42], identifies absorption efficiency and temperature-dependent heat transfer coefficients. Once the calibration is completed, a simulation-based power control optimization is performed using the software `FEMUSS` [36, 37, 39, 40, 43–45], aiming to maintain consistent melt-pool penetration during deposition. The resulting

thermal histories are used to evaluate residual stresses, distortions, and microstructural evolution through the Johnson–Mehl–Avrami–Kolmogorov (JMAK) model [46] and microhardness estimation following [47].

The paper is structured as follows: Section 2 presents the simulation tools for component qualification, including the AM activation strategy, thermo-mechanical coupling, and microstructure modeling. Section 3 introduces the process qualification framework with the power and dwell time control strategies. Section 4 provides calibration and comparative case studies between constant and optimized power conditions, and Section 5 summarizes the main findings and future research directions.

2 Component qualification

In this section, we briefly overview the numerical tools used to assess component qualification, acknowledging that detailed descriptions are available in previous works [37, 39, 40, 43–45]. The AM activation strategy based on G-code is adopted for the deposition sequence of the new added material. An embedded meshing technology is then employed to efficiently generate adaptive, non-conforming meshes for the repaired/remanufactured component, capturing complex geometries with reduced computational overhead. These geometric models feed into a robust thermo-mechanical simulation that accounts for the interplay between thermal gradients and mechanical stresses during the repair process. Additionally, the integration of the Johnson–Mehl–Avrami–Kolmogorov (JMAK) model and a dedicated microhardness model allows us to predict microstructural evolution and correlate it to mechanical performance. Together, these tools enable a comprehensive, non-destructive evaluation of residual stresses, distortions and metallurgical properties, ensuring that the repaired/remanufactured component meets the required quality standards.

2.1 Thermo-mechanical modeling of AM processes

The simulation framework begins by processing the G-code for the AM part, which contains the printing sequence and process parameters. This G-code is read and processed by `FEMUSS` to activate the computational domain corresponding to the deposition zones. The simulation is performed on a regular, unfitted mesh initially composed of two bounding-boxes that enclose the AM part and its substrate. An octree algorithm refines these elements to a predetermined level, ensuring that the Finite Element(FE) analysis accurately captures the geometries involved. Refinement and activation are carried out using a bounding box constructed around the heat source position based on the scanning sequence specified in the G-code. Collision detection between this bounding box and the finite elements identifies those that require refinement and activation [37, 48–50]. This strategy not only ensures a faithful representation of the evolving deposition but also facilitates efficient meshing by updating only the relevant parts of the computational domain.

Once the activated domain is established, the coupled thermo-mechanical problem is addressed using a staggered scheme. The thermal analysis is performed first by solving the heat equation, which captures the rapid heating and cooling cycles inherent in DED processes. The heat source is modeled either by a Gaussian/double ellipsoid distribution [51–54] or by an average power density method [12, 55, 56]. Phase-change effects are neglected due to the timescale of analysis being much larger than that required to capture rapid phase transitions [57]. Then, the computed thermal field informs the mechanical analysis, where small strain kinematics are assumed and an elasto-visco-plastic constitutive model is employed. This model splits the total strain into elastic, thermal and visco-plastic components. Thermal strains are determined using a temperature-dependent expansion coefficient, while visco-plastic strains are derived based on the von Mises yield criterion and the principle of maximum plastic dissipation. For further details on the constitutive modeling, including the treatment of thermal strains and visco-plastic behavior, the reader is referred to [44, 47, 51, 58, 59].

2.2 Embedded approach for the remanufactured component modeling

The embedded approach is integrated with Adaptive Mesh Refinement and coarsening (AMR) to provide a robust and efficient framework for automatically generating FE meshes directly from CAD geometries. This is particularly beneficial when using CAD geometries of the remanufactured or repaired components, such as those obtained from 3D scanning, as the substrate for the subsequent material deposition by AM. AMR plays a pivotal role by dynamically refining the mesh in regions where the geometric accuracy is critical while coarsening it in less essential areas, thereby reducing computational costs significantly. In contrast to traditional meshing techniques that require uniformly fine meshes to capture intricate details, the Embedded-AMR strategy delivers high-fidelity representation in key regions without incurring excessive computational demands.

Remark: For remanufacturing or repair operations, the CAD geometry of the damaged component is typically acquired through 3D scanning, and the corresponding RS-274 (GCode) is generated to restore the nominal geometry. In the present work, since data acquisition of the damaged component was outside the scope of the study, the

provided CAD model is assumed to represent the existing surface condition. The embedded strategy itself is general and can be applied to either damaged or corrected geometries, as it operates independently of the specific CAD input.

A discrete level-set, defined as a signed distance function ϕ, is computed for all mesh nodes as the minimum orthogonal distance to the surface of the embedded geometry. The sign indicates whether a node lies inside (negative) or outside (positive) the solid. The inside-outside classification is determined by casting multiple rays per node and adopting the statistically consistent result, thereby minimizing false detections; points with an even number of intersections lie outside the body, whereas those with an odd number lie inside the CAD geometry. Figure 2 shows the computed signed distance field within the background FE mesh.

Determining the discrete signed distance function enables a straightforward classification of FEs. A FE is classified as *active* if at least one of its nodes has a non-positive signed distance function. Mathematically, this condition is expressed as:

$$\min_{i\in\{1,2,\ldots,N\}} \phi_i \leq 0 \tag{1}$$

where N is the total number of nodes in the FE and ϕ_i denotes the signed distance function at node i. When $\phi_i = 0$, the corresponding node lies exactly on the surface. Under this classification, all FEs either fully inside or intersected by the surface are considered active, thereby avoiding the need for explicit operations on cut elements.

This workflow applies to both NURBS-based (STEP, IGES) and triangulated (STL) geometries. Since direct distance evaluation on complex NURBS surfaces can be computationally expensive, the CAD geometry is first converted to a triangulated representation prior to signed distance evaluation [45]. The procedure uses the Open Cascade Technology kernel [60] and efficient distance-evaluation algorithms based on [61].

Figure 3 shows the entire workflow for the substrate generation later used in Sect. 4 to demonstrate the newly developed method. Figure 3a depicts the geometry in STL format and Fig. 3b and c show the level-set function, including the iso zero surface, representing the boundary of the embedded substrate. Figure 3d shows the final mesh after several levels of mesh refinement. The element classification yields elements entirely inside the embedded body■, elements cut by its boundary■ and elements entirely outside■.

2.3 Microstructure evolution of Ti6Al4V

The evolution of the microstructure during Metal Additive Manufacturing (MAM) processes is critical to define the mechanical properties and performance of AM components. AM processes using titanium alloys, such as Ti6Al4V, exhibit complex thermal cycles that lead to multiple phase transformations during solidification and solid-state transitions. The primary crystal structures include the beta phase (β), a body-centered cubic (BCC) structure stable at high temperatures and the alpha phase (α), a hexagonal close-packed (HCP) structure stable at lower temperatures [46]. At room temperature, Ti6Al4V typically consists of a mixture of α and β phases.

During the cooling of Ti6Al4V, β grains nucleate and grow from the liquid phase. Below the β-transus temperature (T_β, approximately 995–1000 [°C]), the $\beta \rightarrow \alpha + \beta$ transformation occurs, forming Widmanstätten α structures. This diffusion-controlled transformation is often modeled using the JMAK equation [46, 62–64], which relates the

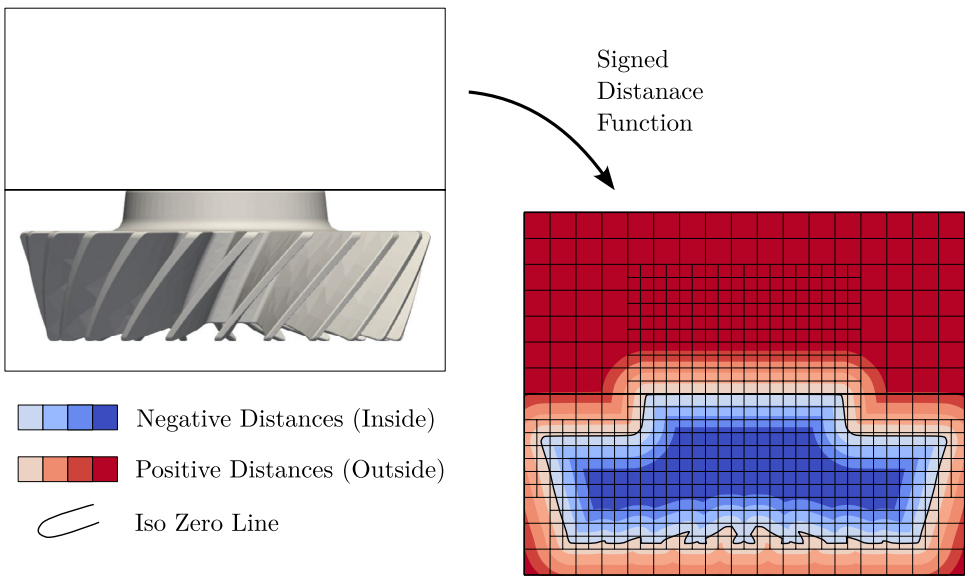
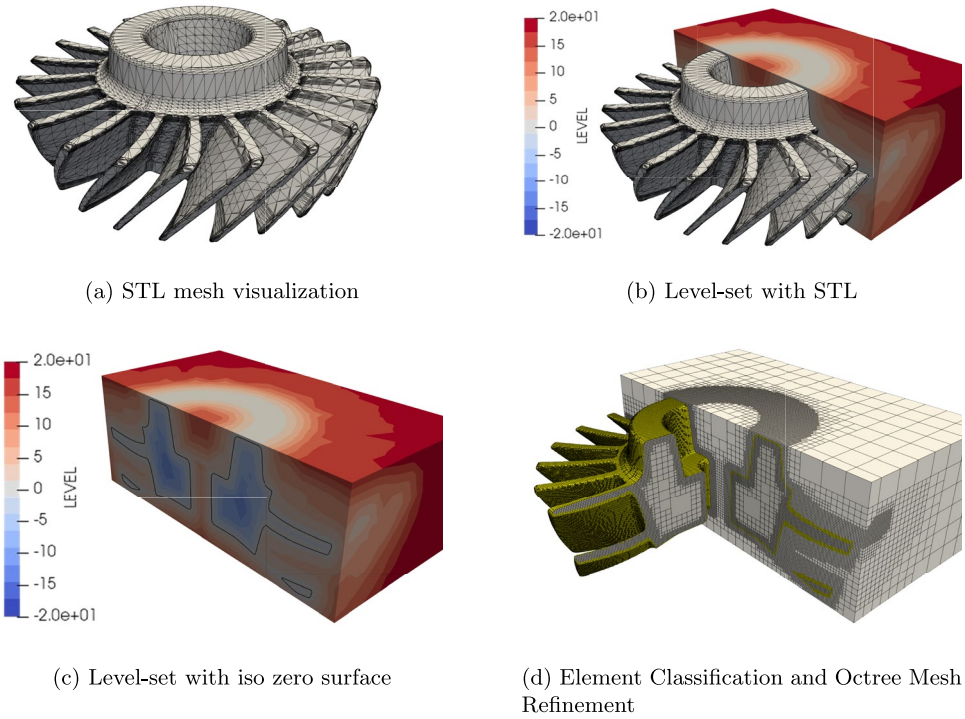


Fig. 2 Computation of singed distance function, used to classify and activate elements of the substrate, that is the component to be remanufactured



(a) STL mesh visualization

(b) Level-set with STL

(c) Level-set with iso zero surface

(d) Element Classification and Octree Mesh Refinement

Fig. 3 CAD file to computational mesh workflow

phase fractions to the Temperature-Time Transformation (TTT) diagrams. Faster cooling rates lead to the formation of martensitic α_{m}, a diffusionless transformation governed by the Koistinen-Marburger equation [65].

The repeated heating and cooling cycles typical of AM processes result in incomplete transformations and complex phase distributions. The high cooling rates can form metastable martensite, which may partially dissolve during subsequent thermal cycles due to heat accumulation. At lower cooling rates, coarser α lamellar structures are formed, often coexisting with martensitic and α-basket weave structures. Prolonged exposure in the transformation range ($T_{\mathrm{diffusion}} \leq T \leq T_{\beta}$) leads to coarsening mechanisms, such as Ostwald ripening and transformation coarsening, driven by α/β interface migration, thus:

1. **Formation of Widmanstätten Alpha** (α_{w}): A diffusion-controlled transformation of $\beta \rightarrow \alpha_{\mathrm{w}}$ during cooling below the β-transus temperature ($T_{\beta\mathrm{trans}}$).
2. **Formation of Martensite** (α_{m}): A diffusionless transformation of $\beta \rightarrow \alpha_{\mathrm{m}}$ during rapid cooling below the martensite start temperature (T_{m}).
3. **Dissolution of Martensite**: Conversion of metastable martensite $\alpha_{\mathrm{m}} \rightarrow \alpha_{\mathrm{w}} + \beta$ during reheating above the martensite dissolution start temperature (T_{mdiss}).
4. **Alpha to Beta Transformation**: Reheating above the α -transus temperature ($T_{\beta\mathrm{start}}$) causes the reverse transformation of α (both α_{w} and α_{m}) into β.

Note that $T_{\mathrm{m}} = (650 - 500\,[^{\circ}C]) \sim T_{\beta\mathrm{start}} = (600 - 650\,[^{\circ}C])$ for Ti6Al4V, T_{m} indicates a cooling cycle and $T_{\beta\mathrm{start}}$ a heating cycle. A thorough explanation on the model is found in reference [46].

The resulting microstructure strongly influences the mechanical properties of Ti6Al4V. While finer microstructures, such as those formed at high cooling rates, enhance tensile strength, defects like porosity and inclusions may reduce ductility. Numerical models, such as those based on JMAK and empirical laws for martensitic transformations, provide a framework for predicting phase evolution and optimizing process parameters to achieve the desired properties.

2.3.1 Microhardness model for Ti6Al4V

This model links the microstructural evolution of DED-fabricated Ti6Al4V with its mechanical performance through Vickers microhardness measurements. It correlates key microstructural features-specifically, the α-lath width-with hardness values as shown in [66]. Thermal cycles during the DED process directly influence the microstructure: prolonged high temperatures lead to coarser α-laths and lower hardness, while rapid cooling results in finer structures and increased hardness.

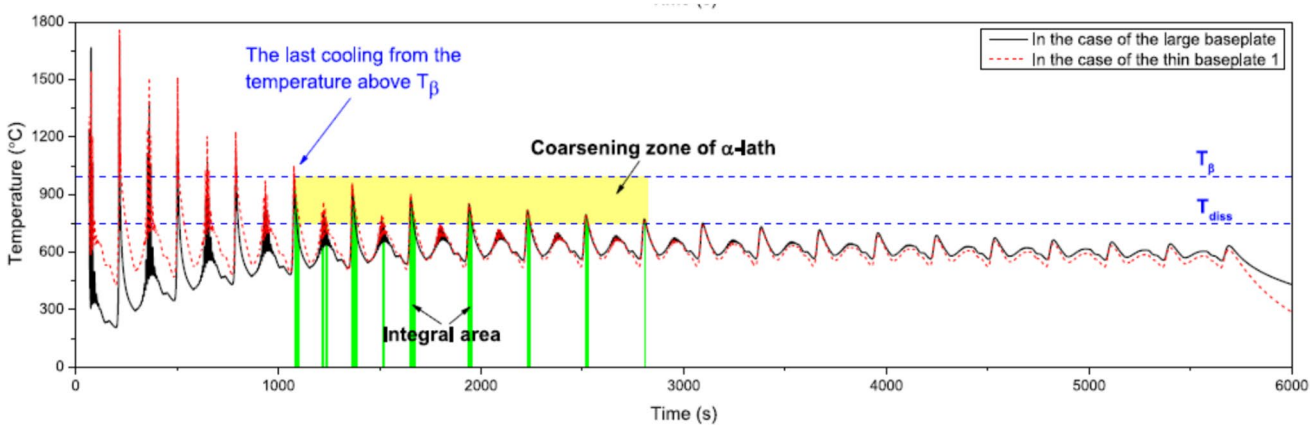


Fig. 4 Temperature plot at a representative point, with the shaded area indicating the cumulative thermal contribution (IA) [47]

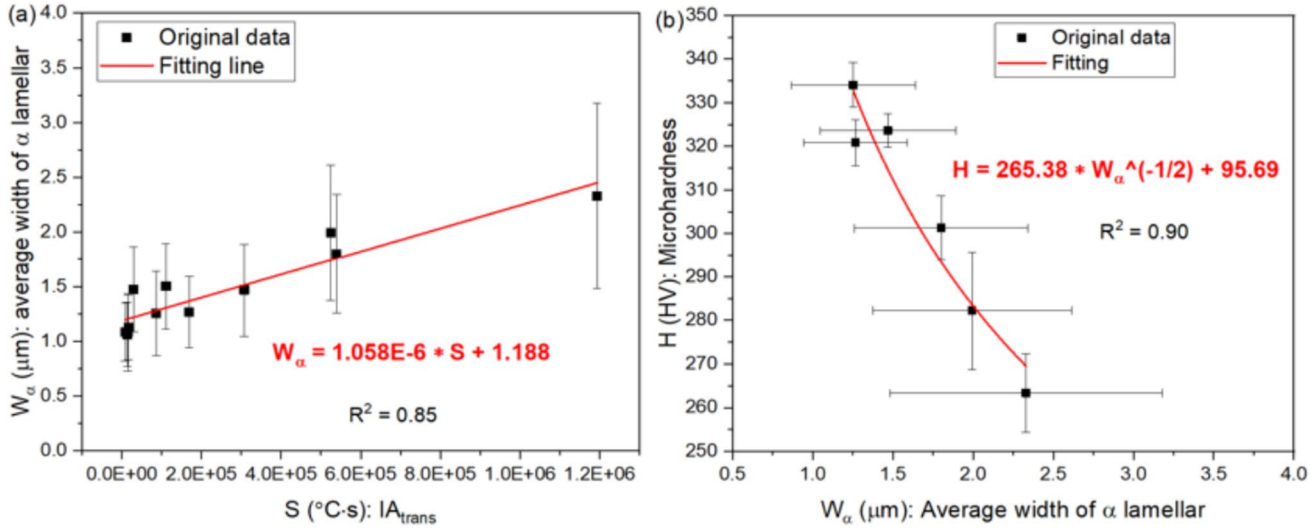


Fig. 5 Experimental relationship curves: **a** Correlation between thermal history and α-lath width; **b** Relationship between α-lath width and Vickers microhardness [66]

To quantify thermal exposure, an Integral Area (IA) sums the time spent between the α-lath dissolution temperature, T_{mdiss} and the β-transus temperature, $T_{\beta\mathrm{trans}}$:

$$IA = \int_{T_{\mathrm{mdiss}}}^{T_{\beta\mathrm{trans}}} T(t)\, dt \tag{2}$$

Figure 4 illustrates the temperature profile at a representative point, with the shaded area corresponding to the IA. A higher IA indicates greater thermal exposure, leading to coarser α-laths and reduced microhardness, whereas a lower IA promotes finer microstructures and higher hardness.

The relationship between IA and the α-lath width is derived from experimental observations reported in [66], which established an empirical correlation for Ti–6Al–4V as:

$$W_\alpha = 1.058 \times 10^{-6} \times IA + 1.188 \tag{3}$$

where W_α is measured in μm. The α-lath width, in turn, correlates with the Vickers microhardness, H, as:

$$H = 265.38 \times W_\alpha^{-\frac{1}{2}} + 95.69 \tag{4}$$

Figure 5 presents the experimental correlation between the thermal history, α-lath width and Vickers microhardness. Thus, by simulating thermal histories and calculating IA values, the model enables prediction and tailoring of the mechanical properties through adjustments in process parameters such as energy density and cooling rates.

3 Process qualification

Process qualification in DED aims to ensure that the manufacturing process is reliable, repeatable and capable of producing high-quality parts. Unlike component qualification,

which focuses on evaluating the performance of the final part, process qualification examines the parameters and conditions that govern the material deposition itself. This includes aspects such as melt-pool stability, energy input, deposition speed and dwell times, which are critical to maintaining consistent layer quality and minimizing defects like porosity, warping and incomplete fusion. By optimizing these parameters, process qualification enhances buildability, ensuring that the additive process can be successfully completed under varying conditions, whether for repair, remanufacturing, or cladding applications. This section discusses the numerical strategies used to assess and optimize process parameters, emphasizing their impact on deposition continuity and overall process efficiency.

3.1 Power control strategy

The power control strategy is designed to dynamically adjust the power supply to keep the melt-pool penetration with a target constant value. This control is achieved through an iterative loop, where the power input is continuously modified based on the difference between the observed penetration and the target value. The target penetration is typically the penetration value observed in a single track test when it reaches a steady-state condition. Implementing this strategy requires two key stages: (1) accurately computing the morphology of the melt-pool, i.e. its length, width and penetration depth and (2) applying power corrections to achieve the desired penetration.

3.1.1 Computation of melt-pool morphology

The characterization of the melt-pool involves identifying and analyzing its discrete representation and its boundary. This analysis provides its length, width and penetration depth. To determine the melt-pool region, FEs are categorized into two groups based on nodal temperature:

Fully Melted Elements (FME): Elements where all nodes have a temperature above the melting temperature. These elements belong entirely to the melt-pool.

Melt-pool Boundary Elements (MBE): Elements where at least one node has a temperature greater than the melting temperature, but not all nodes exceed this threshold, thus belonging to the melt-pool boundary.

The FMEs are treated as standard FE and the MBEs are populated with an increased number of Gauss points for a more precise characterization of the melt-pool boundary. Each MBE is enriched with 5 Gauss points per dimension, ensuring that the melt-pool boundary is captured with high fidelity. Figure 6 shows the melt-pool and the MBEs in green with increased Gauss points.

The computation of melt-pool dimensions-length (L), width (W) and penetration depth (D)-is carried out within a local coordinate system centered to the moving heat source. The local axes are defined as follows: ξ represents the scanning direction (longitudinal) axis; φ is the transverse direction, perpendicular to ξ in the Cartesian $x-y$ plane; and ζ corresponds to the vertical (z-axis) direction.

The length of the melt-pool, L, is defined as the maximum distance between any two points inside the melt-pool along the scanning direction, ξ. Mathematically, it is expressed as:

$$L = \max_{\mathrm{r}_i, \mathrm{r}_j} \|\mathrm{r}_i - \mathrm{r}_j\|_\xi \tag{5}$$

where

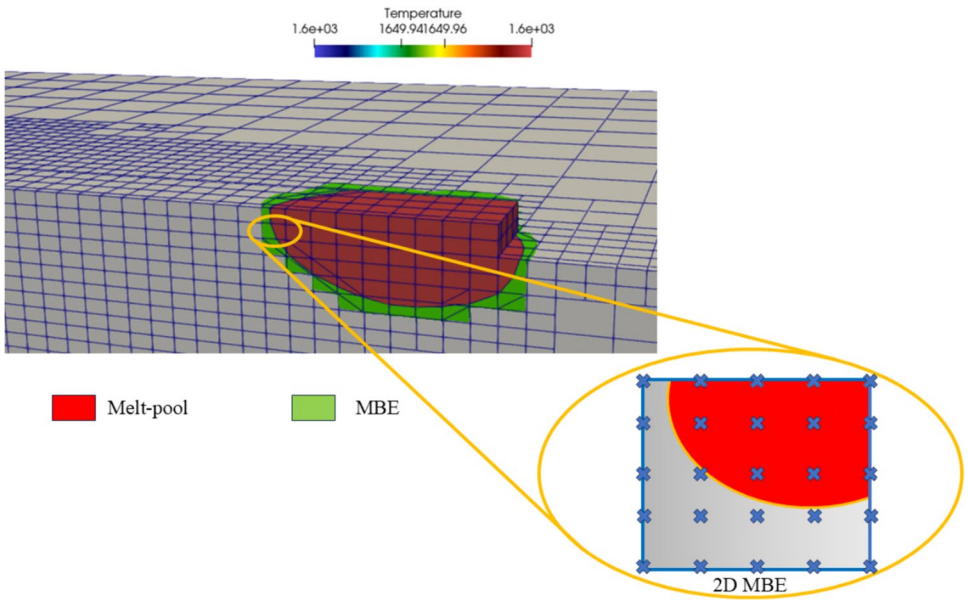


Fig. 6 Numerical computation of the melt-pool length

$$\|\mathrm{r}_i - \mathrm{r}_j\|_\xi = |(\mathrm{r}_i - \mathrm{r}_j) \cdot \hat{\xi}| \tag{6}$$

and $\hat{\xi}$ is the unit vector along the ξ-axis. Figure 7 shows a schematic representation of the melt-pool length computation.

The width, W, represents the maximum distance between points in the transverse direction, φ, which is orthogonal to ξ and lies in the Cartesian x-y plane. This is given by:

$$W = \max_{\mathrm{r}_i, \mathrm{r}_j} \|\mathrm{r}_i - \mathrm{r}_j\|_\varphi \tag{7}$$

where

$$\|\mathrm{r}_i - \mathrm{r}_j\|_\varphi = |(\mathrm{r}_i - \mathrm{r}_j) \cdot \hat{\varphi}| \tag{8}$$

and $\hat{\varphi}$ is the unit vector along the φ-axis.

The penetration depth, D, is computed as the vertical extent of the melt-pool along the ζ-axis. It is determined by the difference between the maximum and minimum ζ-coordinates of points within the melt-pool without accounting for the deposited layer height, h_{layer}:

$$D = \max(\zeta_i) - \min(\zeta_j) - h_{layer} \tag{9}$$

where ζ_i denotes the z-coordinates of the Gauss points in the melt-pool.

The volume of the melt-pool, V, is calculated as the sum of the volumes of the FMEs and the partial contributions from the MBEs where the temperature exceeds the melting threshold. This can be expressed as:

$$V = \sum_{e \in \mathrm{FME}} V_e + \sum_{e \in \mathrm{MBE}} \mathcal{F} \cdot V_e \tag{10}$$

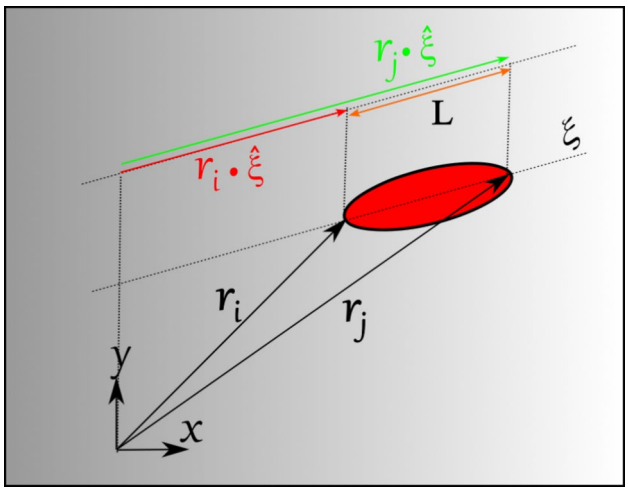


Fig. 7 Numerical computation of the melt-pool length, L

where V_e represents the volume of a fully melted element. The term $\mathcal{F} \cdot V_e$ accounts for the fractional contribution of a MBE.

3.1.2 Power correction strategy

The power correction strategy dynamically adjusts the power supply to ensure that the melt-pool penetration matches the user-defined target value, D_{target}. This adjustment is achieved by continuously monitoring the observed penetration depth, D_{obs} and modifying the power input based on the discrepancy between D_{obs} and D_{target}. The goal is to iteratively minimize the penetration error, $e = D_{\mathrm{target}} - D_{\mathrm{obs}}$, until the system converges to the desired state.

The corrected power for the next iteration, P_{new}, is computed using a linear correction defined as:

$$P_{\mathrm{new}} = P_{\mathrm{current}} \cdot K_p \tag{11}$$

where K_p is the correction factor. Note that K_p is bounded to avoid instability providing smoother adjustments as:

$$K_{pmin} < K_p = 1 + \frac{D_{\mathrm{target}} - D_{\mathrm{obs}}}{D_{\mathrm{target}}} < K_{pmax} \tag{12}$$

where K_{pmin} and K_{pmax} are user-defined lower and upper bounds.

To illustrate the effect of bounding the correction factor, Fig. 8 compares the penetration evolution during the optimization process with and without the K_p limits. In case (a), the absence of bounds results in oscillatory corrections and unstable convergence, while in case (b), applying the bounded K_p ensures smooth and stable convergence toward the target penetration depth in four iterations, confirming the importance of constraining the correction factor within appropriate limits.

The implementation of this strategy follows an iterative workflow. At each time-step, the observed penetration depth, D_{obs}, is calculated and the relative error is determined as $e = (D_{\mathrm{target}} - D_{\mathrm{obs}})/D_{\mathrm{target}}$. The corrected power input is then written into a new GCode file, maintaining the same scanning sequence but updating the process window. This process is repeated iteratively to progressively reduce the penetration error.

By dynamically adapting the power input it is possible to ensure a constant melt-pool morphology independently of the substrate temperature or boundary conditions (heat losses). This strategy is also expected to stabilize layer height and deposition rate by limiting excessive melting and maintaining melt-pool consistency, thereby improving dimensional uniformity. Moreover, the more controlled thermal history associated with power modulation is anticipated

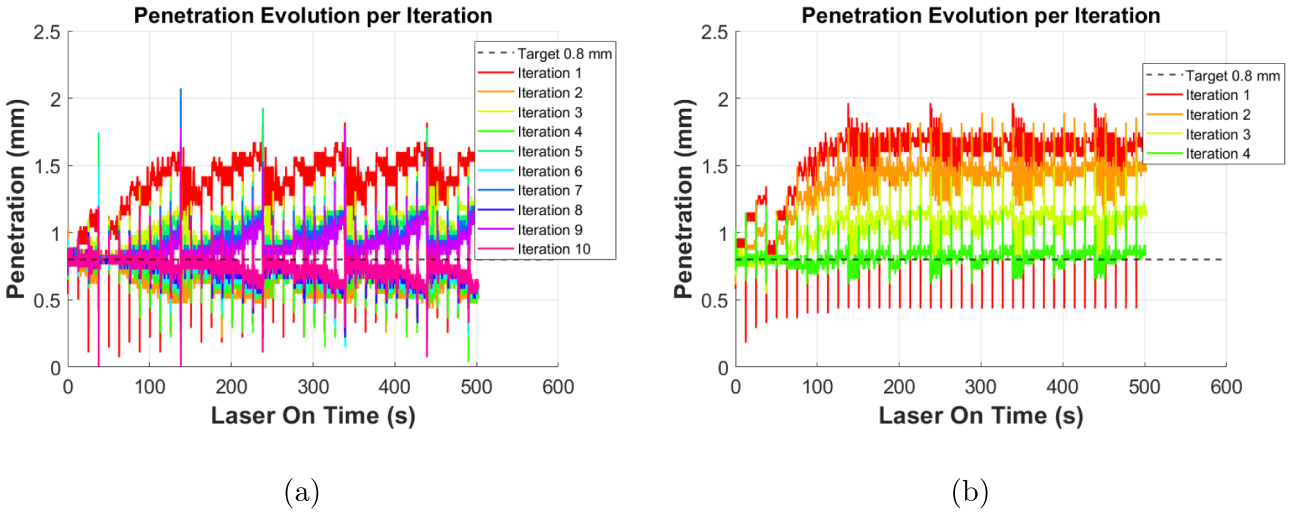


Fig. 8 Penetration evolution per iteration. **a** Without bounding the correction factor K_p, showing unstable convergence behavior. **b** Converged penetration using the bounded correction factor K_p

to favor finer grain structures and a more uniform distribution of α and β phases.

Remark: The power control strategy presented here focuses on controlling the melt-pool penetration depth. However, the approach is general and can be easily extended to other control parameters, such as melt-pool temperature, or volume.

3.2 Power modulation

Power modulation is a necessary step to ensure stability to the power input during the additive manufacturing process. Many 3D printers struggle to adapt to rapid power fluctuations, particularly when frequent adjustments are required. To address this, the power is modulated only at hatch level. This ensures a smooth power modulation with an average control over melt-pool morphology.

Power modulation can be applied at two levels. Hatch-level modulation averages the power across individual hatches, ensuring a constant power input within each hatch while allowing variations between hatches. Layer-level modulation, on the other hand, averages the power over an entire layer, resulting in uniform power across all hatches within the layer. The choice of modulation level depends on the specific application and the capabilities of the 3D printer. Hatch-level modulation offers finer control, whereas layer-level modulation provides greater smoothing and stability.

Let $P_{\text{opt},i}$ be the converged optimized power for the *i*-th segment and t_i be the time required to deposit that segment. The modulated power is then defined as:

$$P_{\text{mod}} = \frac{\sum_{i=1}^{N} (P_{\text{opt},i} \cdot t_i)}{\sum_{i=1}^{N} t_i} \tag{13}$$

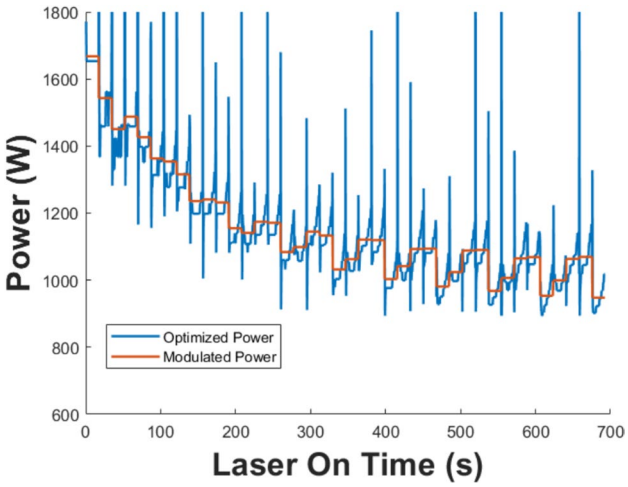


Fig. 9 Optimized power from the correction process versus modulated hatch power

where N is the total number of segments in the hatch or layer. This formula ensures that the average power reflects both the corrected power values weighted by the time spent depositing each segment. Figure 9 shows the converged P_{opt} and the resulting P_{mod}.

3.3 Dwell time control

Dwell time control is an essential strategy for enhancing buildability during the DED process by allowing sufficient cooling between the deposition of successive layers. This approach helps manage the thermal history of the build, thereby mitigating excessive heat accumulation that could lead to uncontrolled melt-pool expansion, distortions and microstructural degradation. The dwell phase in this study is regulated by a target temperature threshold: the next layer is

only activated and deposited once the built component cools to a specified maximum temperature. While this temperature-based approach allows precise thermal control-made possible by the use of pyrometers-it is worth noting that in many practical applications, a fixed dwell time is typically used due to the absence of real-time temperature monitoring. This highlights a key advantage of simulation-based process control, which can optimize dwell times based on predicted thermal behavior, enabling more effective process control even in systems that lack direct temperature sensing capability.

To efficiently simulate the cooling process, a variable time-stepping scheme is employed during the dwell phase. The time step size is dynamically adjusted based on the computed cooling rate and the proximity of the current temperature to the target threshold. When the cooling rate is slow, larger time steps are used to accelerate the simulation, whereas smaller time steps are applied as the temperature approaches the threshold to ensure precise control. By maintaining a consistent temperature threshold before activating the next layer, this dwell time allows for using a more uniform process window as optimized for single-track depositions. Thus, improving overall process reliability.

Figure 10 presents the workflow integrating power correction and dwell time control. The inner loop (in blue) represents the execution of the FE simulation, which computes the local error, applies the bounded correction factor K_p, updates the power for the next iteration $P(i+1)$, and, when enabled, performs the dwell time evaluation for thermal regulation. The outer loop (in orange) is managed by an external script that verifies global convergence and initiates additional iterations when necessary. It is important to note that dwell time control is user-defined, allowing the user to specify whether temperature control should be applied after every layer, after a defined number of layers, or omitted entirely.

4 Numerical examples

This section presents a set of numerical examples designed to evaluate the proposed modeling framework for the qualification of repair and remanufacturing operations by DED of Ti6Al4V components. The section is divided into two main parts. First, a calibration phase is conducted using experimental data to determine key model parameters. Then, the performance of the calibrated model is validated against a set of experimental cases and extended to an industrial remanufacturing case.

The calibration phase employs a 10-layer multi-track deposition (experimental setting C1), where thermocouple data are used to calibrate the power absorption efficiency parameter and the heat loss at the boundaries defined in terms of temperature-dependent Heat Transfer Coefficients (HTCs). The calibration process is carried out using the `DAKOTA` software [42], which facilitates automated parameter estimation by minimizing the least-squares error difference between the simulated results and thermocouple data. The resulting calibrated values are then validated against additional experimental datasets, which have variations in either process parameters or geometrical features with respect to C1.

The first set of numerical examples evaluates buildability aspects to different substrate heights (5 $[\mathrm{mm}]$, 15 $[\mathrm{mm}]$ and 30 $[\mathrm{mm}]$) and compares the impact of using constant power versus modulated power. Key metrics analyzed include dimensional accuracy, residual stress distribution and microstructural evolution. This example highlights how substrate height variations influence thermal gradients and, consequently, the mechanical and microstructural integrity of the remanufactured part.

The next example focuses on dimensional accuracy and microstructure evolution, comparing results obtained using a constant power supply with those achieved through modulated power control. The analysis aims to demonstrate how power modulation helps to mitigate issues such as excessive warpage and distortion while increasing microstructural homogeneity.

In the third example the combined effects of power modulation and dwell time control are evaluated. The dwell time control, based on a target temperature threshold between layers, allows the build to cool down, enhancing dimensional accuracy, stabilizing microstructure evolution and reducing residual stress concentrations. The comparison is made between three configurations: constant power, modulated power and modulated power with dwell time control.

The final example demonstrates the practical applicability of the modeling framework to a complex industrial remanufacturing scenario: the customization of a rotor with an extended axle. In this case dimensional accuracy, residual stresses and microstructure evolution in a real-world remanufacturing process is examined. The analysis underscores the versatility of the framework in addressing intricate geometries and highlights its effectiveness in ensuring high-quality, qualified remanufactured components.

The GCodes for the first three numerical examples were generated manually to ensure precise control of the scanning sequence and deposition parameters. For the remanufacturing case, the toolpath was generated using the Ultimaker Cura slicer software [67].

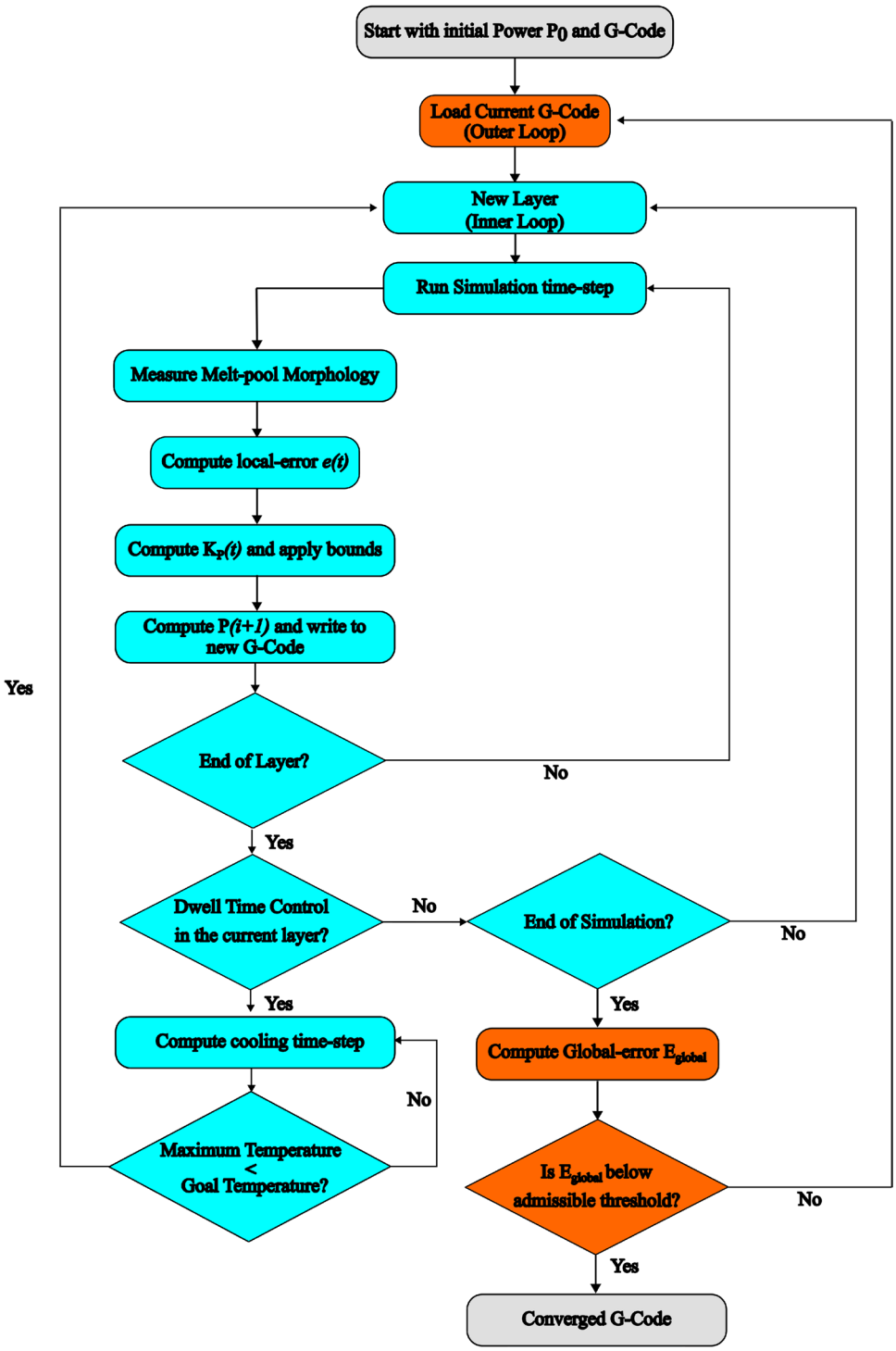


Fig. 10 Power correction and dwell time control workflow

4.1 Multi-track example

The first scenario consists of different rectangular substrates clamped to one free end. The AM consists of 10 layers metal deposition of 4 parallel tracks, carried out using a powder DED system equipped with a 10 [kW] fibre laser at the MIIT Key Laboratory of Metal High Performance Additive Manufacturing and Innovative Design in the Northwestern Polytechnical University. The deposition is performed in a sealed chamber filled with argon to prevent oxidation during laser cladding. Temperature-dependent thermo-mechanical material properties for Ti6Al4V are sourced from [68] and the JMAK model from [46]. According to reference [68], a heat transfer coefficient (HTC) of 50 $[\mathrm{W/m^2\,^\circ C}]$ accounts for the heat loss of the substrate in the clamped region.

Figure 11 illustrates the experimental setup, showing the positioning of the thermocouples (TC1, TC2 and TC3), the virtual displacement control point (DC), the scanning sequence for both even and odd layers, as well as the printed specimens (C1–C4). A laser spot size of $5\,[\mathrm{mm}]$ and a spacing of $3\,[\mathrm{mm}]$ between scan vectors were employed. Powder was delivered at a rate of $6\,[\mathrm{g/min}]$, with a gas flow rate of $6\,[\mathrm{L/min}]$. Each demonstrator is characterized by a unique combination of power, scanning speed, layer thickness and substrate size, as summarized in Table 1.

The calibration process involves fitting the temperature measured at TC1 during experiment C1. This is performed by optimizing a temperature-dependent HTC and the absorption efficiency ϖ. The calibrated value for the absorption efficiency is $\varpi = 24.75\,\%$ and Table 2 presents the calibrated HTC values.

The in-situ temperature is measured using Omega GG-K-30 type K thermocouples with a measurement uncertainty of $2.2\,[^\circ\mathrm{C}]$ or 7.5%. The thermocouples signals are recorded using a Graphtec GL-900 8 high-speed data-logger.

The discrete domain is generated using mesh adaptivity, following the framework described in [37, 39, 40, 43].The FEs of the substrate have a size of $2\,[\mathrm{mm}]$ in the $X-Y$ plane, while those of the component are refined to $0.5\,[\mathrm{mm}]$. Additionally, each layer is discretized into 2 FEs in the Z direction.

The comparative temperature profiles depicted in Fig. 12 demonstrate the calibrated model's effectiveness in capturing the transient thermal behavior across four multi-track deposition settings (C1–C4). Figure 12a–c corresponds to experiment C1 used for the calibration of the simulation. Although thermocouple TC2 failed during printing, the simulation shows excellent agreement with the experimental results up to the point of failure, while the TC1 and TC3 measurements align very closely with the simulation data.

The calibrated parameters were further validated on settings C2, C3 and C4, which feature variations in power density and substrate size. In C2, a higher power density yields elevated peak temperatures and the simulation accurately reflects these increases, as shown in Fig. 12d–f. In contrast, the thicker substrates in C3 ($15\,[\mathrm{mm}]$) and C4 ($30\,[\mathrm{mm}]$) provide enhanced heat dissipation, resulting in lower temperature gradients that are also well captured by the simulation, as evidenced in Fig. 12g–l.

The deviations observed in Fig. 12g, h, and j are mainly attributed to experimental and modeling factors. In cases (h) and (j), the differences result from thermocouples whose welds failed during deposition, leading to unstable or inaccurate temperature signals. For Fig. 12g, the deviation can be explained by either an underestimated HTC at the clamping interface, maintained at 50 $[\mathrm{W/m^2\,^\circ C}]$ from previous tests on thinner substrates, or by a possible misplacement of the thermocouple closer to the clamp. Both scenarios justify the initial agreement between the experimental and simulated curves, followed by the divergence observed at later stages, when the clamping region warms up and begins to dissipate heat more effectively. It is worth noting that the thermocouple located farther from the clamping region (TC3) in experiments C3, Fig. 12i, and C4, Fig. 12l, shows very good agreement between experimental and simulated temperature profiles, confirming that the thermal model accurately reproduces the temperature evolution away from boundary-dominated regions.

4.1.1 Modulated power assessment

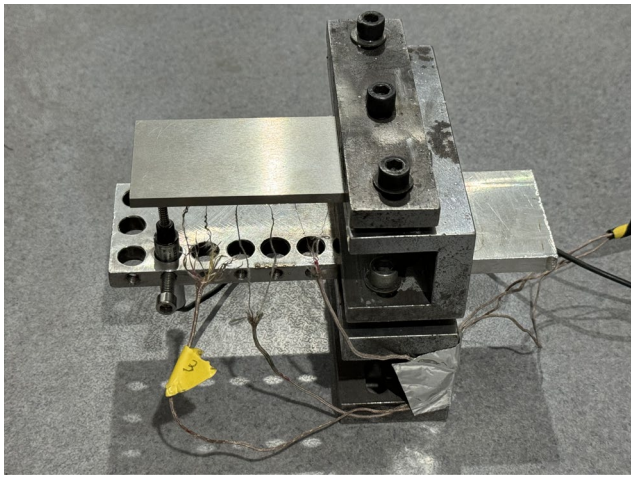
Maintaining the melt-pool at a controlled size is critical for achieving geometrical accuracy [20]. The modulated power levels were determined using the procedure outlined in Sect. 3.1.2, targeting a penetration depth of $0.85\,[\mathrm{mm}]$. This value is derived from the analysis of the first layer of each experiment once steady-state conditions are achieved, assuming this as the reference process window of the DED machine (refer to Fig. 13a, which illustrates the constant power penetration during the steady-state phase of the first hatch).

Figure 13a shows penetration over time for three power modes: constant, optimized and modulated. Achieving optimal penetration requires continuous power adjustment, a capability not supported by all printers. The modulated power approach minimizes abrupt power fluctuations while maintaining penetration within the target range by assuming a constant power supply for each hatch. Figure 13b compares the constant, optimized and modulated power profiles for experiment C1, highlighting their respective performances in achieving the target penetration.

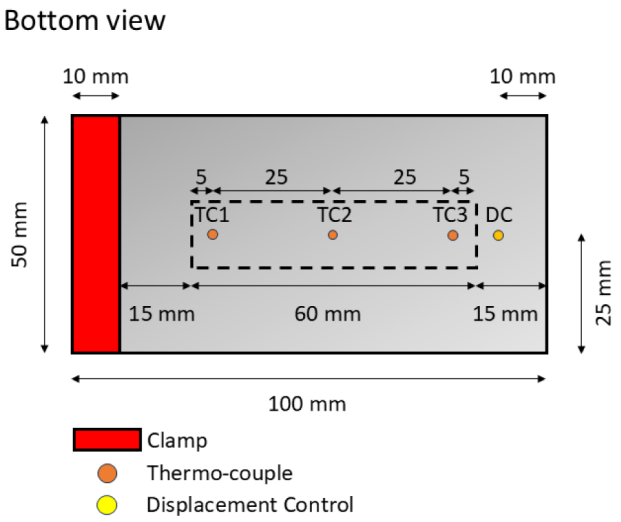
4.1.2 Thermal-response analysis

The thermal history of AM processes plays a critical role in determining the microstructural and mechanical properties of the final component. This subsection examines the impact of power control on thermal evolution and cooling rates across different substrate conditions, as observed in experiments C1 through C4.

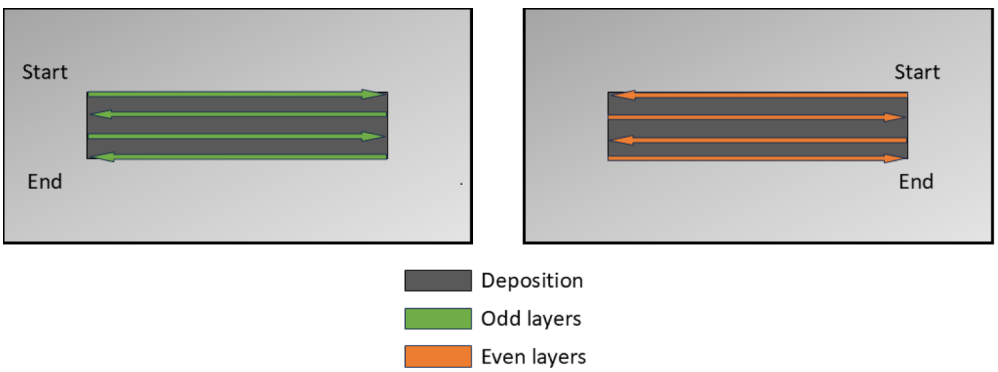
To simplify the analysis and avoid redundancy, the conclusions drawn from condition C1 apply equally to conditions C2 through C4. Although all experiments exhibit similar trends and support comparable interpretations, only the results for C1 and C4 are presented in detail here, as they represent the extremes values for the substrate thickness. This approach ensures clarity and focus, while maintaining the validity of the findings across all experimental conditions.



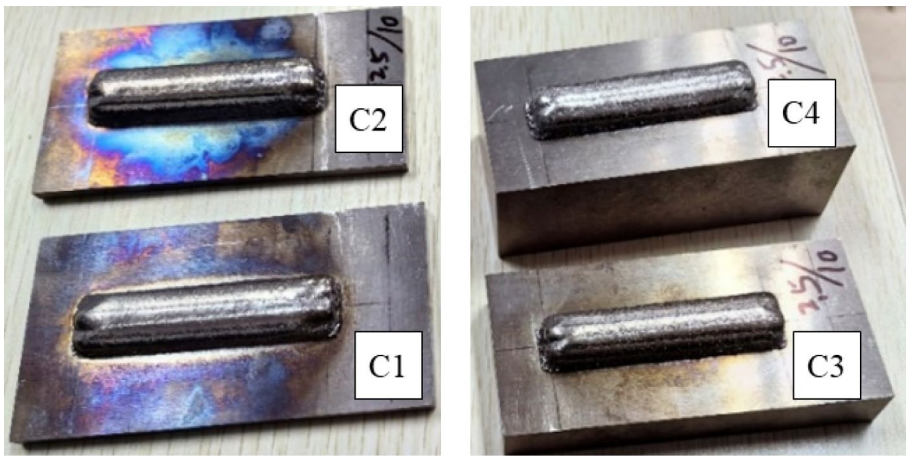
(a)



(b)



(c)



(d)

Fig. 11 Multi-track: **a** Experimental setup. **b** Substrate geometry with thermocouple and displacement control positions. **c** Scanning sequence per layer. **d** Printed specimens C1–C4

Table 1 Multi-track process parameter and geometry of the demonstrators

No.	Power (P)	Scann. speed (v)	Back speed	P/v	Layers	Subst. size	Layer thick.
	[kW]	[mm/s]	[mm/s]			[mm^3]	[mm]
C1	3	15	15	0.20	10	100 × 50 × 5	0.5
C2	2.5	10	15	0.25	10	100 × 50 × 5	0.7
C3	2.5	10	15	0.25	10	100 × 50 × 15	0.7
C4	2.5	10	15	0.25	10	100 × 50 × 30	0.6

Table 2 Multi-track calibrated temperature-dependent HTC values

Temperature ($^\circ C$)	HTC ($W/m^2\,^\circ C$)
20	1.27
600	35.17
1200	120.45
1600	1170.54
2000	1170.54

Figure 14 shows the temperature evolution recorded by thermocouples TC1–TC3 for the modulated and constant power setups in experiment C1, along with the comparison of cooling rates. As expected, the constant power configuration yields higher temperature profiles, greater fluctuations and increased cooling rates.

Figure 15 presents similar trends for experiment C4 regarding the power strategies. However, the thicker substrate with higher thermal inertia produces smaller, steadier temperature increases compared to the oscillatory behavior observed in C1. Consequently, the recorded cooling rate during deposition is null and the higher heat capacity of the thicker substrate results in extremely low cooling rate values at the thermocouple locations.

High temperatures during deposition can vaporize materials or lead to gas entrapment, which increases porosity and compromises structural integrity [19]. Prolonged exposure to elevated temperatures, especially in the presence of oxygen, may cause surface oxidation, significantly reducing the lifespan of the component [69]. Heat accumulation can destabilize the process, leading to inconsistent layer deposition and defective parts.

The cooling rate in MAM plays a critical role in determining the microstructure and mechanical properties of fabricated components. Faster cooling rates, driven by rapid solidification, generally promote finer microstructures, which can enhance mechanical strength [19]. However, these higher cooling rates can have drawbacks, such as reduce ductility, trap gas pockets, or cause incomplete fusion, leading to porosity and other defects that compromise part integrity [70–72].

4.1.3 Mechanical-response analysis

The mechanical response was evaluated by monitoring displacement evolution at a control point (DC) and analyzing residual stress profiles along critical sections of the substrate. In experiment C1–C4, as shown in Fig. 16, the modulated power setup produces slightly lower peak displacements in the deposition direction (X-axis) and negligible differences in the transverse (Y) direction. In the build (Z) direction, modulated power induces larger displacements due to higher thermal gradients; however, the vertical movement at the control point remains more or less stable around $0.75\,[\mathrm{mm}]$ during the deposition for C1 and C2, ensuring a consistent distance between the laser and the substrate and thereby improving geometric accuracy.

Conversely, the displacements in the building direction for modulated and constant powers of experiments C3 and C4 result in lower differences compared to C1 and C2. Thicker or larger substrates, such as in C3 and C4, absorb and dissipate heat more effectively. The heat-sinking effect of thicker substrates can homogenize thermal gradients distribution and reduce sharp fluctuations in displacement, regardless of the power mode adopted. Meanwhile, thinner or smaller substrates, such as in C1 and C2, are more sensitive to rapid temperature changes, which can amplify the displacement differences when using modulated power.

The residual stress distributions in Fig. 17 reveal distinct differences between the modulated and constant power strategies. Under constant power, the heat input is uniform, leading to deeper thermal penetration into the substrate and lower in-plane gradients. By contrast, the modulated power approach focuses the heat input more locally, leading to increased stress concentrations near the component-substrate interface while reducing residual stresses in the deeper regions of the substrate, particularly in thinner substrates (see Fig. 17a–d).

Substrate thickness significantly influences the observed stress patterns. In cases C1 and C2, where the substrate is relatively thin ($5\,[\mathrm{mm}]$), the modulated power strategy results in higher local temperature gradients due to limited thermal inertia, intensifying stress concentrations at the interface. Conversely, in cases C3 and C4, the thicker substrates ($15\,[\mathrm{mm}]$ and $30\,[\mathrm{mm}]$, respectively) offer greater heat-sinking capacity, which reduces the temperature gradient differences between constant and modulated power configurations at the substrate. However, the increased stiffness of these thicker substrates also contributes to higher residual stresses in the component, as the reduced deformability increases

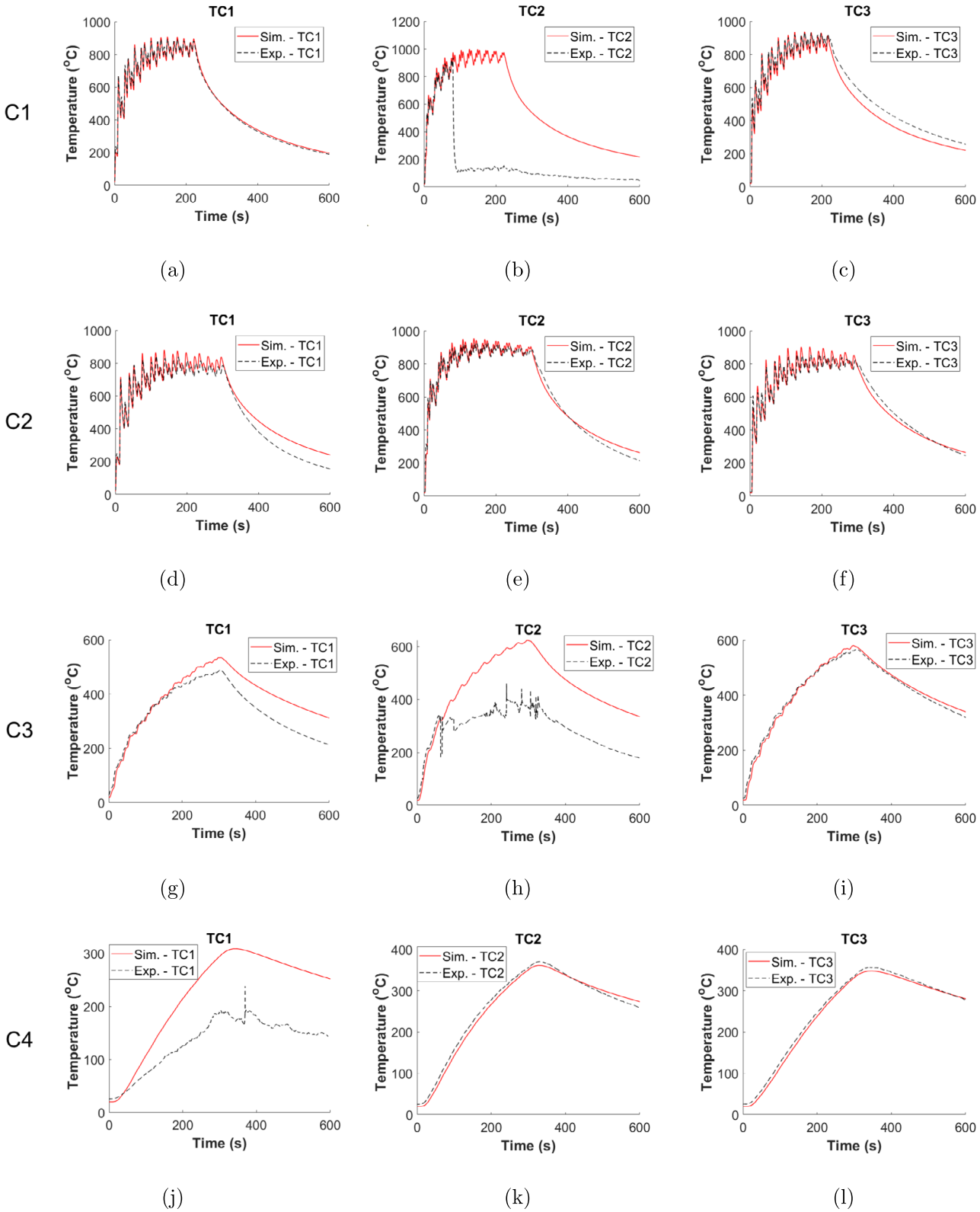


Fig. 12 Multi-track: experiment C1 **a**–**c** TC1–TC3, respectively. Experiment C2 **d**–**f** TC1–TC3, respectively. Experiment C3 **g**–**i** TC1–TC3, respectively. Experiment C4 **j**–**l** TC1–TC3, respectively

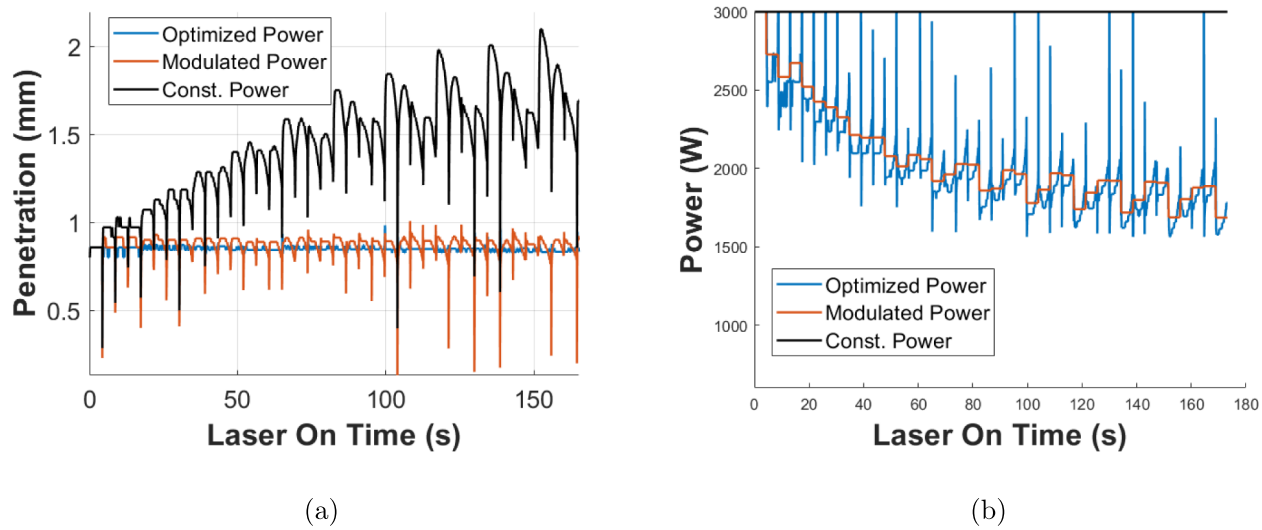


Fig. 13 Multi-track: experiment C1 **a** Penetration values for the optimized, modulated and constant power. **b** Power curves for the optimized, modulated and constant power

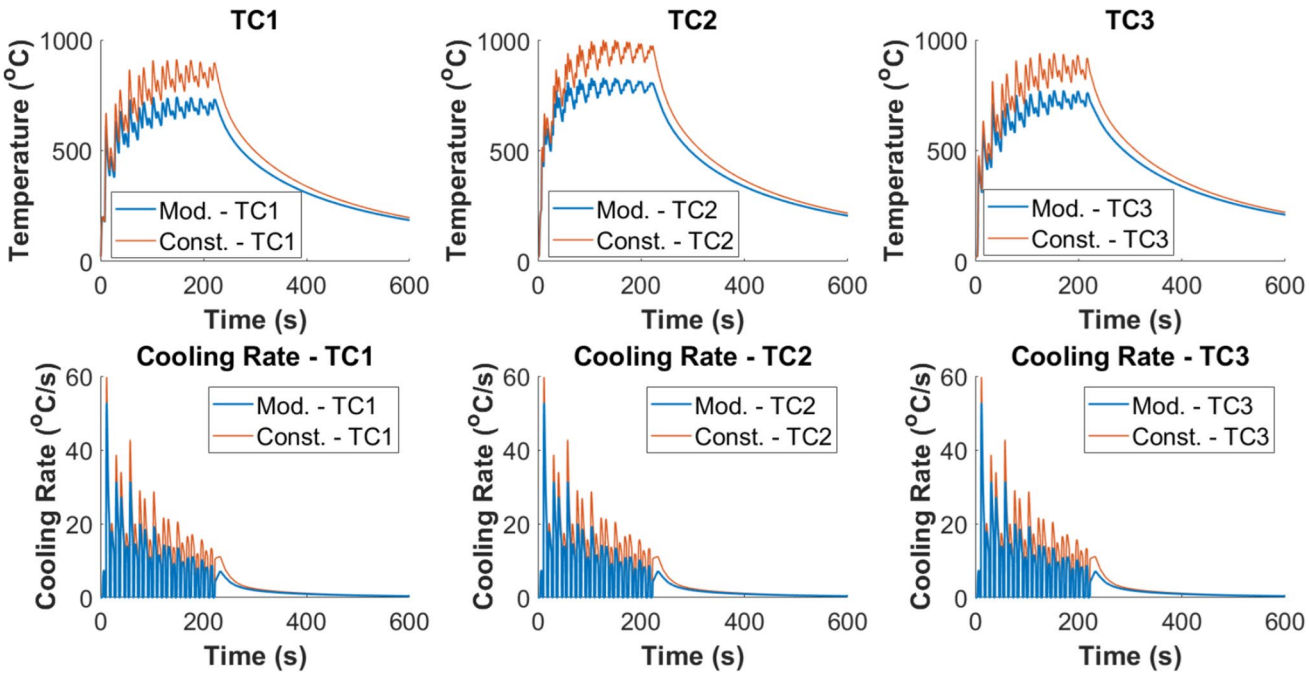


Fig. 14 Multi-track analysis: C1 experiment. The top row illustrates the temperature evolution recorded by thermocouples TC1–TC3, comparing modulated and constant power setups. The bottom row depicts the corresponding cooling rates observed for TC1–TC3 under both power configurations

stress accumulation. Moreover, the lower heat input from modulated power can still lead to elevated residual stresses at the component level due to the pronounced thermal gradients between layers. This effect is clearly illustrated in the residual stress fields shown in Fig. 17a–d, where C1 and C2 display comparatively lower internal stress magnitudes, while C3 and C4 exhibit pronounced residual stresses in the build direction. The stiffer boundary conditions imposed by the thicker substrates restrict the thermal contraction of the deposited layers, limiting the potential for stress relaxation through substrate deformability. As a result, the accumulated thermal strains during cooling promote the buildup of residual stresses within the deposit.

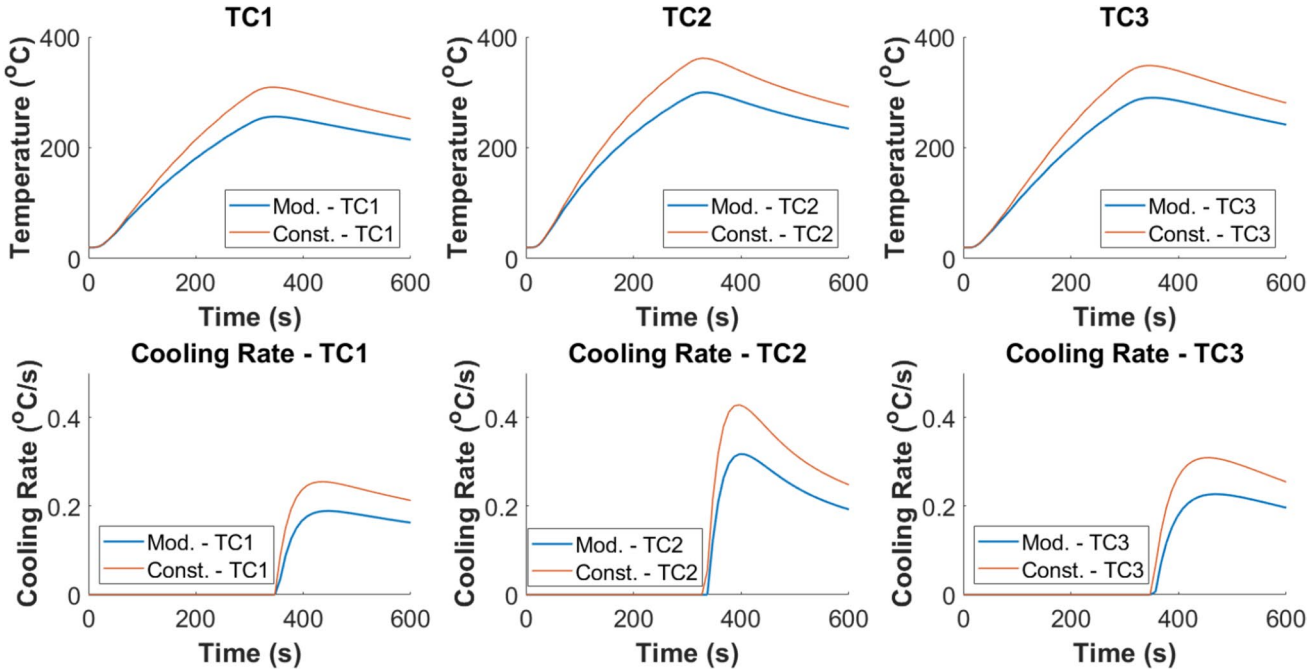


Fig. 15 Multi-track analysis: C4 experiment. The top row illustrates the temperature evolution recorded by thermocouples TC1–TC3, comparing modulated and constant power setups. The bottom row depicts the corresponding cooling rates observed for TC1–TC3 under both power configurations

4.1.4 Microstructure analysis

This subsection analyzes the microstructure evolution under constant and modulated power through a detailed graphical and statistical comparison. The study evaluates key microstructural features, including α_{M}, α_{W} and β fractions, as well as Vickers microhardness. For simplicity and to avoid redundancy, only the results for the extreme cases, i.e. experiments C1 and C4, are presented.

Enhancing microstructural homogeneity improves the uniformity of mechanical properties by minimizing local variability, thereby promoting more predictable performance under service conditions and reducing the likelihood of localized stress concentrations. While an increased martensite fraction (α_{M}) can enhance strength and wear resistance, it may also compromise ductility. Similarly, higher microhardness values contribute to improved surface durability and load-bearing capacity but can render the material more brittle and challenging to machine. A reduction in the beta phase generally results in a harder, stronger microstructure with better wear resistance; however, it may also lower ductility and diminish the capacity of the material to absorb impact energy. Moreover, an excessive amount of β can lead to microstructural instability or undesirable phase transformations, particularly during subsequent heat treatment or in-service thermal cycling. Ultimately, the optimal qualification of the microstructure depends on the targeted microstructural feature and the most effective process control strategy is intrinsically linked to the type of substrate on which the deposition is performed.

Figure 18 illustrates the microstructural evolution observed in the C1 experiment, which was conducted on a thinner substrate ($5\,[\mathrm{mm}]$) with a slightly lower power density. The top row corresponds to the modulated power case, while the bottom row depicts the constant power case. The presented plots represent the distributions of α_{M}, α_{W}, β and microhardness.

Under modulated power, α_{M} exhibits a more uniform distribution at the component level, accompanied by consistent microhardness values across both the substrate and the component. Notably, no β phase is present within the first three layers of the component, while higher β concentrations appear in the subsequent layers. In contrast, the constant power mode results in increased α_{W} content and a lower β fraction in the uppermost layers, while the β phase remains more concentrated in the region near the substrate-component interface. Additionally, the constant power mode leads to broader microhardness variations, indicating greater heterogeneity in mechanical properties.

Figure 19 presents the microstructural results for the C4 experiment, where a thicker substrate ($30\,[\mathrm{mm}]$) and a higher power density are employed. In this configuration, the differences between the power modes become more pronounced at the component level while remaining relatively negligible in the substrate. Specifically, the modulated power mode enhances the homogeneity of both the phase

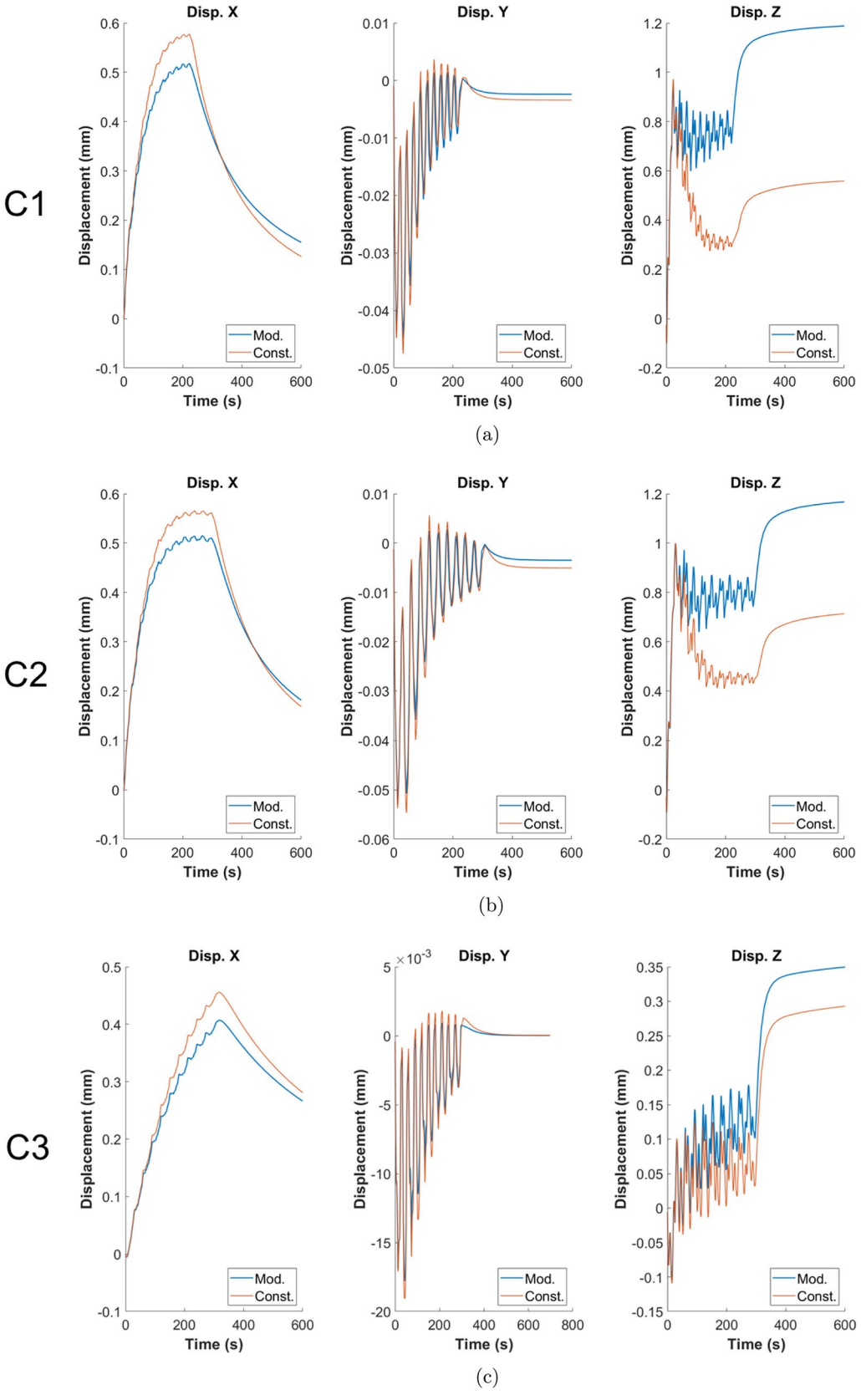


Fig. 16 Multi-track: displacement evolution comparison between the constant and modulated power at the control point DC. Subfigures **a** corresponds to displacements X, Y and Z of experiment C1; **b** to C2; **c** to C3; and **d** to C4

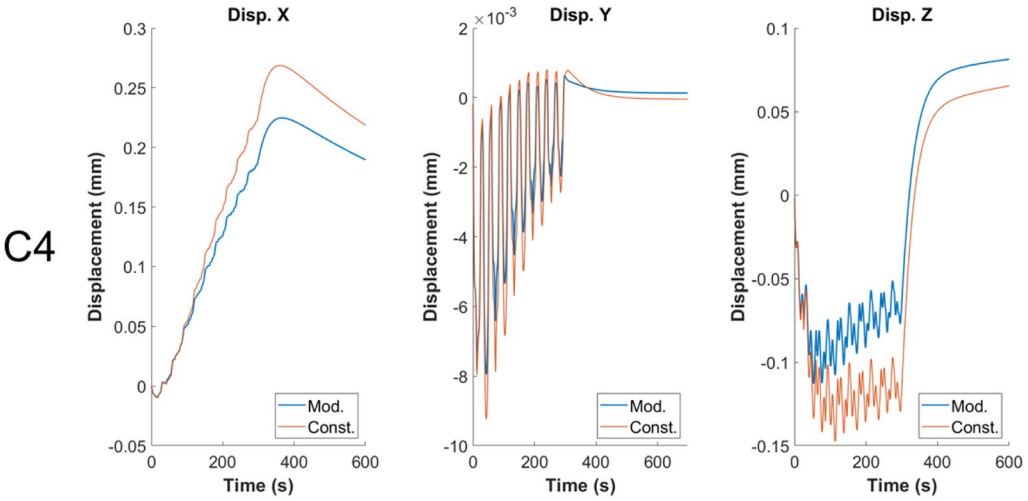


Fig. 16 (continued)

distribution and microhardness, leading to marked increases in α_{M} content and a concomitant rise in microhardness, along with a notable reduction in the β fraction. The robust heat-sinking capacity of the thicker substrate confines the extent of microstructural evolution to a limited region around the component and the substrate-component interface compared to the thinner substrate. Nonetheless, under constant power conditions, local variations in α_{M} and α_{W} concentrations, as well as in microhardness values, become more pronounced in conjunction with elevated β levels, underscoring that non-modulated input can yield sharper microstructural variations in especially larger builds.

4.2 Single-wall example

In the second scenario, a single-wall deposition is investigated under identical clamping conditions and with the same DED blown powder system described in Sect. 4.1. The experiment was conducted at the MIIT Key Laboratory of Metal High-Performance Additive Manufacturing and Innovative Design, located at Northwestern Polytechnical University. The build is performed on a substrate measuring $110 \times 56 \times 6\,[\mathrm{mm}^3]$, with a single-wall of dimensions $80 \times 5 \times 20\,[\mathrm{mm}^3]$ deposited sequentially over 30 layers with a deposition speed of $10\,[\mathrm{mm/s}]$ and laser spot-size of $5\,[\mathrm{mm}]$, see Fig. 20. This configuration enables a detailed evaluation of dimensional accuracy and microstructural evolution in a simplified geometry, thereby complementing the insights from the multi-track experiments. Two experimental configurations are compared: one using a constant power supply, and the other employing an optimized, modulated power control strategy. The analysis focuses on assessing both the geometrical fidelity and the microstructural characteristics of the deposited wall.

4.2.1 Modulated power assessment

Figure 21 illustrates the optimized power curve implemented in the modulated power supply configuration. This curve was derived using the procedure described in Sect. 3.2 and Sect. 4.1, specifically to mitigate melt-pool instability during deposition. By contrast, the constant power configuration maintains a fixed input of $3\,[\mathrm{kW}]$ throughout processing. Figure 22 presents melt-pool visualizations at three representative stages of the deposition process for both power strategies. The results show that the constant power setup leads to uncontrollable increase of the melt-pool volume, affecting both buildability and performance of the build.

4.2.2 Dimensional accuracy

Dimensional accuracy was assessed by comparing the geometry of the deposited walls. Figure 23 illustrates this comparison, with Fig. 23a corresponding to the modulated power setup and Fig. 23b to the constant power setup. The wall deposited under constant power exhibited significant geometric inaccuracies-specifically, an increasing wall width along the build direction and non-uniform layer thickness. In contrast, the modulated power approach yielded a geometry that closely conforms to the nominal design, demonstrating superior dimensional accuracy. In addition, Fig. 23b clearly shows increased oxidation, i.e. evidenced by the distinctive coloration, attributable to the higher temperatures experienced during the constant power process.

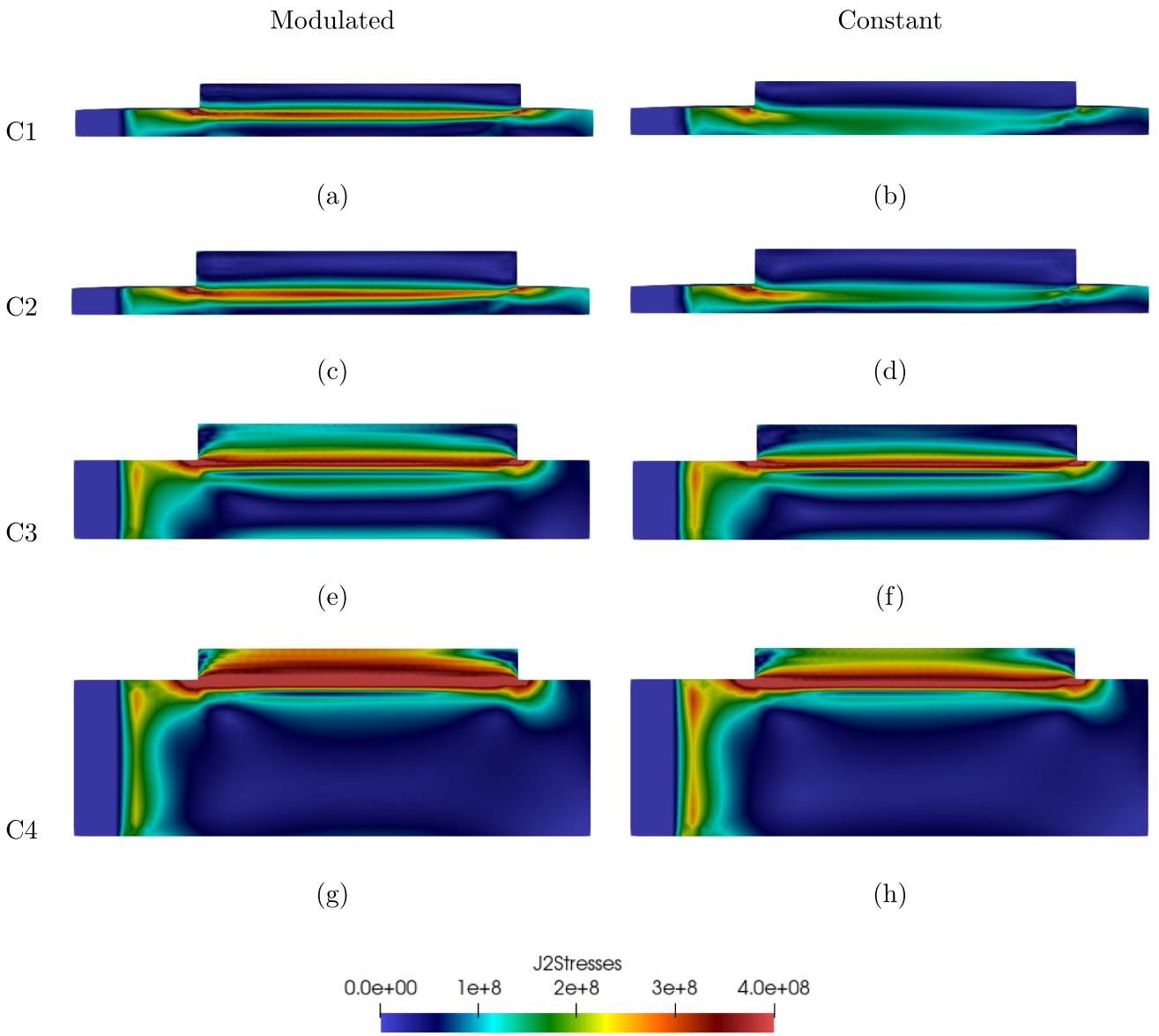


Fig. 17 Multi-track residual stress comparison across the substrate-component cross section. Subfigures **a** and **b** correspond to experiment C1 with modulated and constant power, respectively; **c** and **d** to C2; **e** and **f** to C3; and **g** and **h** to C4

4.2.3 Microstructure analysis

In addition to the geometric evaluation, the microstructure of the deposited walls was investigated to assess metallurgical homogeneity. Comparative microhardness data obtained from simulation, Fig. 24, and experimental grain-size analyses, Fig. 25, are provided for both the constant power and modulated power approaches. Notably, the simulation hardness results serve as an indirect measure of grain size, with higher hardness values indicating finer grains and lower hardness values corresponding to coarser grains. Analysis of the simulated hardness reveals that, in the constant power setup, the coarsest grain region is observed at the bottom of the wall, whereas for the modulated power approach, the coarsest grains are located at mid-height. These findings are corroborated by the experimental assessment of the grain structure.

In summary, the numerical example demonstrates that the modulated power supply significantly enhances both the dimensional accuracy and the microstructural homogeneity of the single-wall, as confirmed by the experimental results. These improvements not only mitigate common issues such as inconsistent layer height and distortion observed with constant power settings, but also promote a more uniform microstructure by reducing variations in phase distribution. This enhanced uniformity is critical for ensuring the performance and durability of remanufactured and repaired

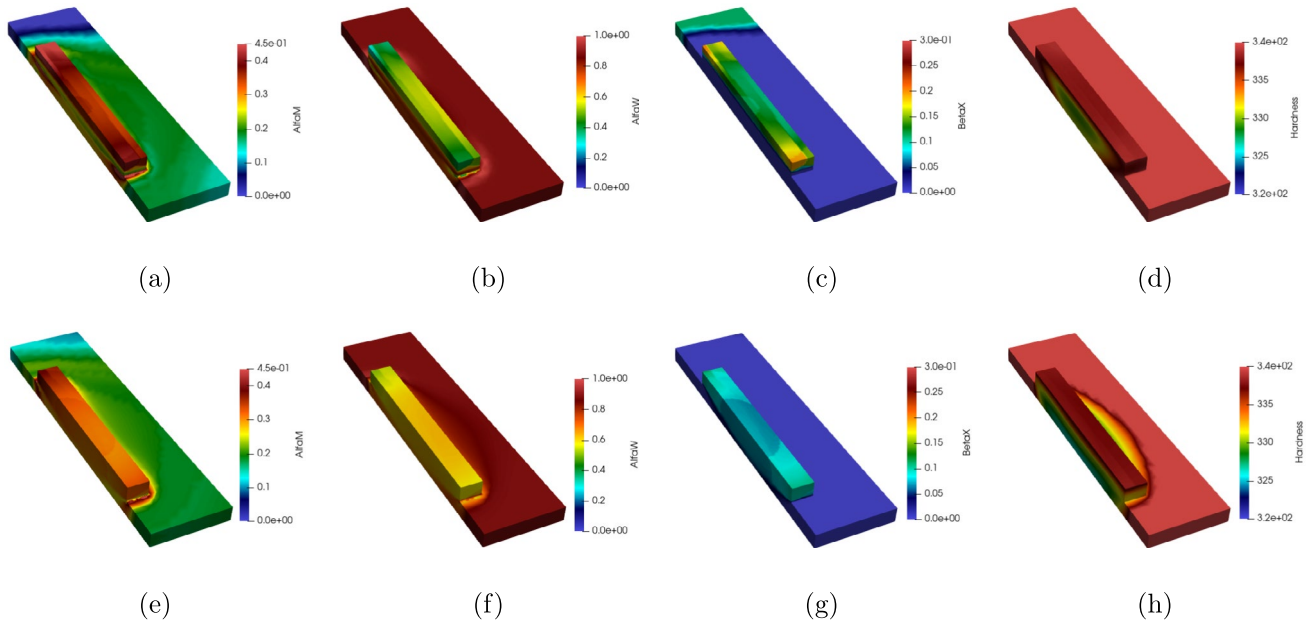


Fig. 18 Multi-track: Experiment C1. Subfigures **a**–**d** show the microstructural parameters under modulated power conditions-specifically, the fractions of α_{M}, α_{W}, β and the measured microhardness, respectively. Subfigures **e**–**h** display the corresponding results under constant power mode

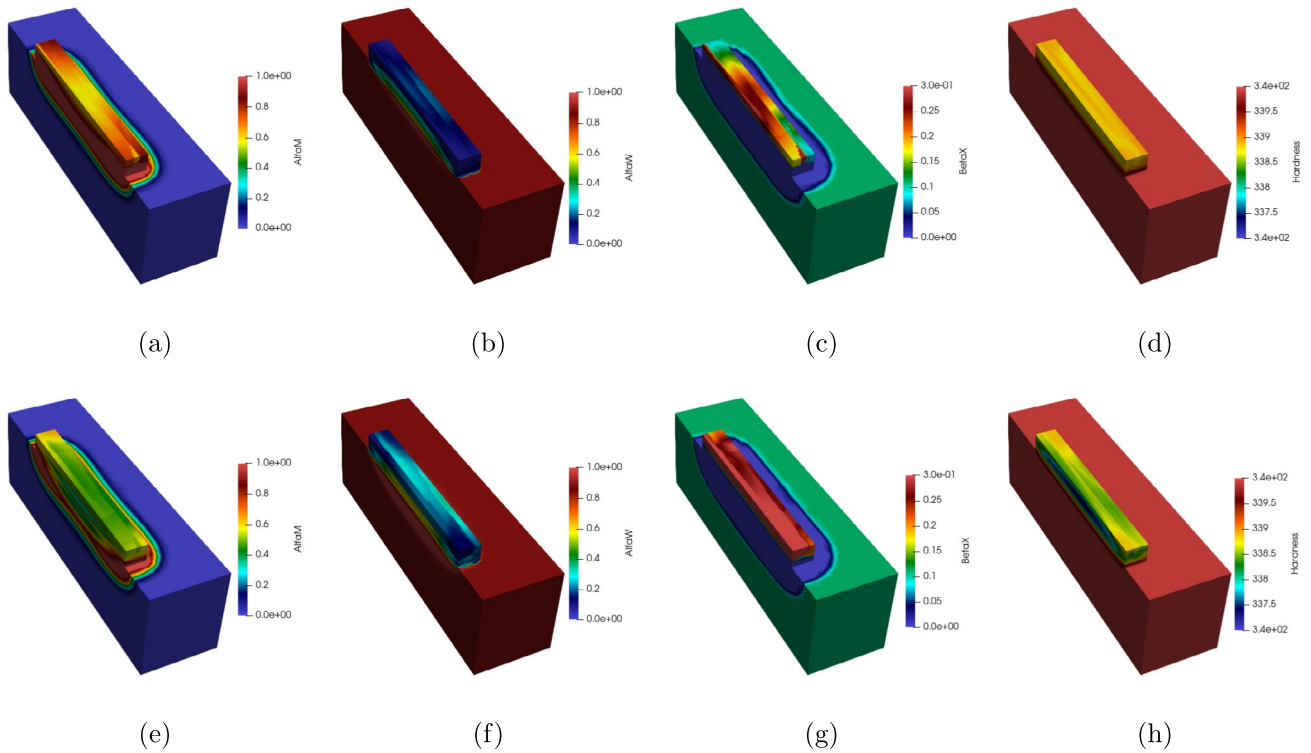


Fig. 19 Multi-track: Experiment C4. Subfigures **a**–**d** show the microstructural parameters under modulated power conditions-specifically, the fractions of α_{M}, α_{W}, β and the measured microhardness, respectively. Subfigures **e**–**h** display the corresponding results under constant power mode

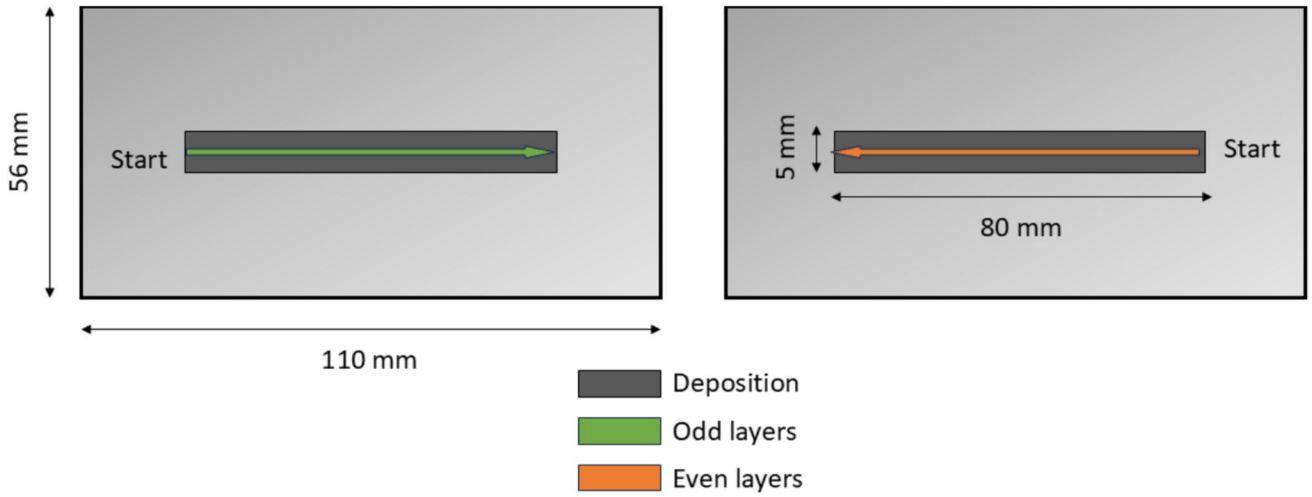
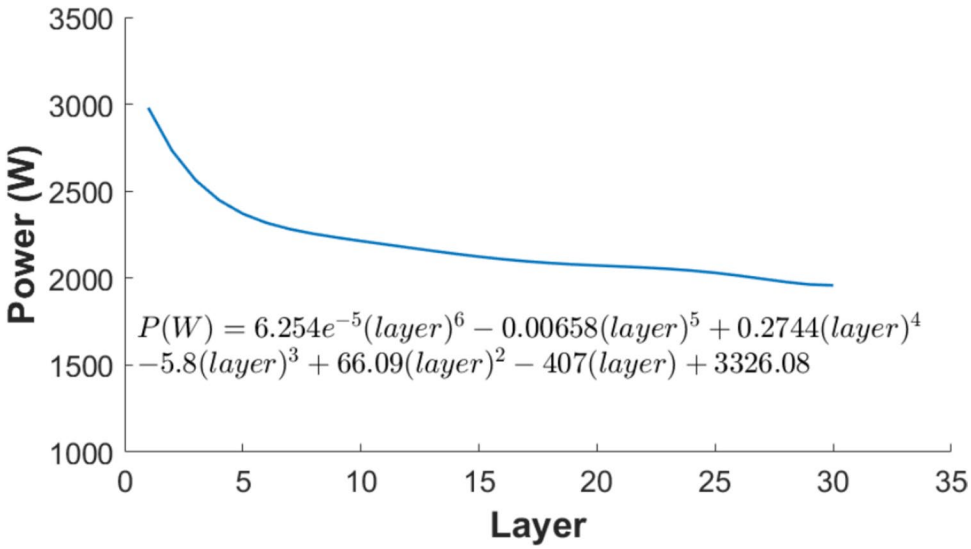


Fig. 20 Single-wall: Scanning sequence per layer and geometry

Fig. 21 Single-wall: Modulated power profile per layer



components, paving the way for more reliable additive manufacturing processes.

4.3 Double-wall example

In this third example, the combined effects of power modulation and dwell time control are evaluated using a TRUMPF TruLaser Cell 7020 system equipped with the DepositionLine package for automatic laser metal deposition with experiment conducted at the Centre for Additive Manufacturing established at the RMIT University. The beam is generated by the TruDisk 3001 laser module, delivered via a dual-clad fibre optic cable, and focused onto the workpiece by MSO 70 welding optics.

It is worth noting that both the DED equipment and the boundary conditions in this example differ from those in previous cases. The thermal calibration of the new experimental conditions was performed using a smaller reference geometry, a thin-walled cylinder as the calibration procedure involves multiple iterations to determine the most suitable temperature-dependent HTC, see Fig. 26a. Here, the substrate of $100 \times 50 \times 12\,[\mathrm{mm}^3]$ is supported by two refractory blocks rather than clamped at one end. The thermocouple setup is presented in Fig. 26b, where thermocouple C, used for the calibration, is embedded at mid-thickness of the substrate. The calibrated temperature evolution curve is shown in Fig. 26c. In addition, the calibrated temperature-dependent HTC used in this study is presented in Table 3 and the calibrated absorption efficiency is $\varpi = 35.56\,\%$.

In this setup, a double-wall deposition was produced, consisting of two parallel tracks with final dimensions of $100 \times 4 \times 50\,[\mathrm{mm}^3]$, deposited over a substrate of dimensions $140 \times 50 \times 12\,[\mathrm{mm}^3]$ and supported by two refractory blocks as in the calibration process. Deposition is carried out within an argon-filled chamber to minimize oxidation, see Fig. 27a. The build comprises 200 layers

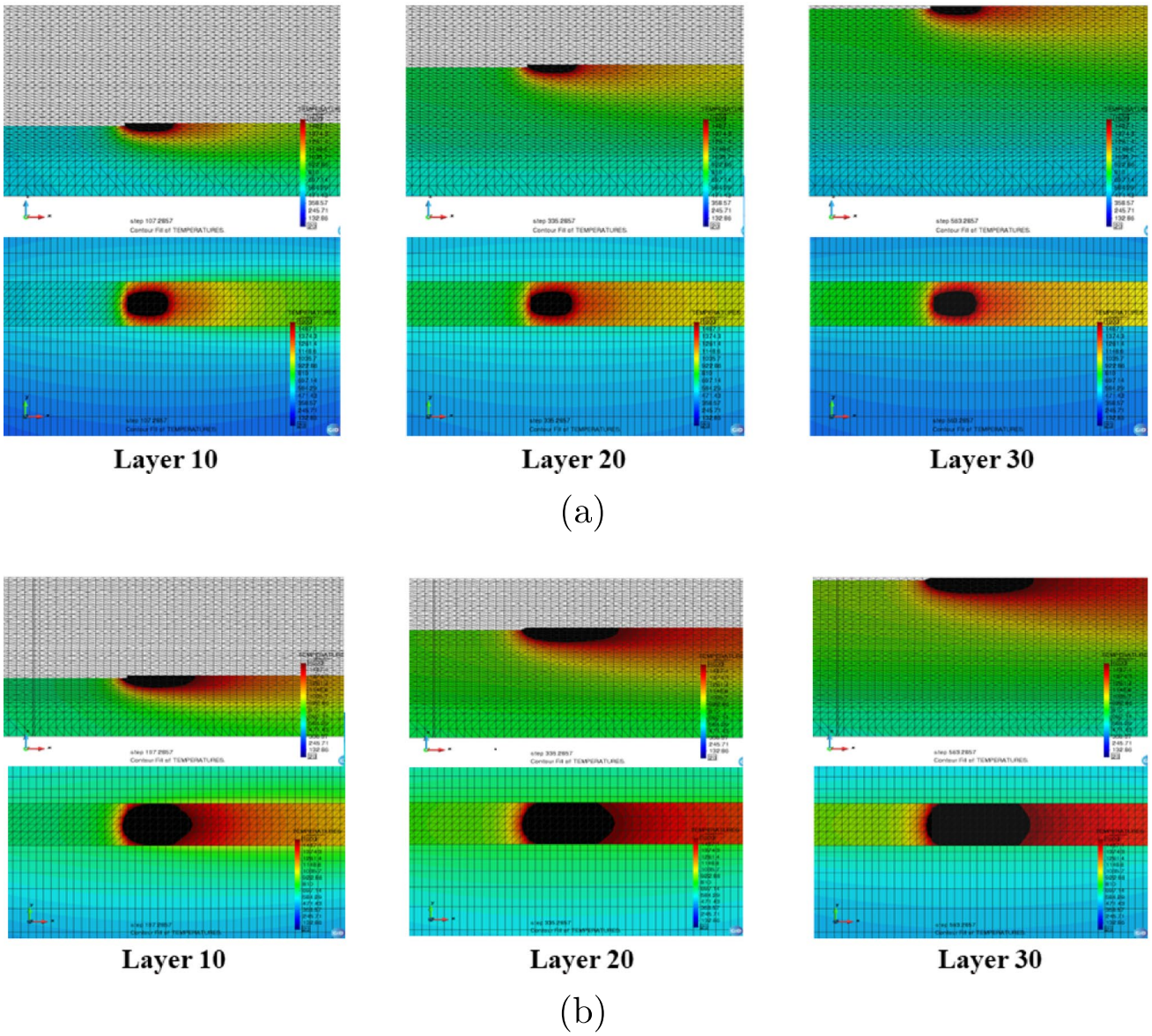


Fig. 22 Single-wall: Melt-pool visualization at three different layers. **a** Modulated power. **b** Constant power

deposited at a scanning speed of $25\,[\mathrm{mm/s}]$. A laser spot size of $3.5\,[\mathrm{mm}]$ and a spacing of $1.5\,[\mathrm{mm}]$ between scan vectors were employed. Powder was delivered at a rate of $5.4\,[\mathrm{g/min}]$, with helium as the carrier gas at $10\,[\mathrm{L/min}]$ and argon as the shielding gas at $16\,[\mathrm{L/min}]$. Pre- and post-acceleration distances of $10\,[\mathrm{mm}]$ were applied to ensure the nozzle maintained a constant speed during wall deposition, see Fig. 27b.

Three experimental configurations are compared: (I) constant power, (II) modulated power and (III) modulated power with dwell time control. The dwell time control is activated every 5 layers, pausing deposition until the maximum temperature drops below $500\,[^{\circ}\mathrm{C}]$, enabling sufficient cooling between layers.

4.3.1 Power and dwell time assessment

Figure 28 displays the optimized power profile used for both the modulated power and controlled inter-layer cooling configurations, along with the inter-layer pause intervals applied every five layers. It is noteworthy that the TRUMPF machine restricts to a maximum of eight power supply levels, thereby impeding per-layer modulation and requiring that the modulation remain within these constraints. In contrast to the constant power setup, which sustains a constant laser output, the modulated approach dynamically adjusts the limited set of power values to maintain a stable melt-pool. Simultaneously, the dwell time ensures that the

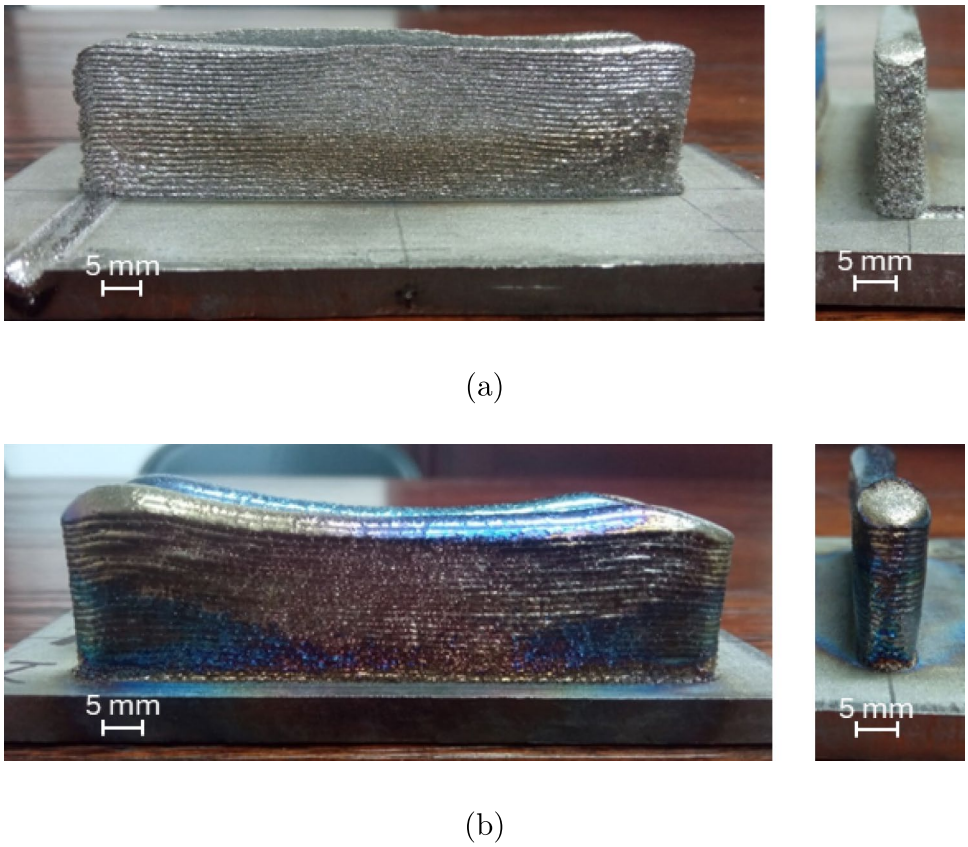


Fig. 23 Single-wall: Dimensional accuracy assessment. **a** Modulated power. **b** Constant power

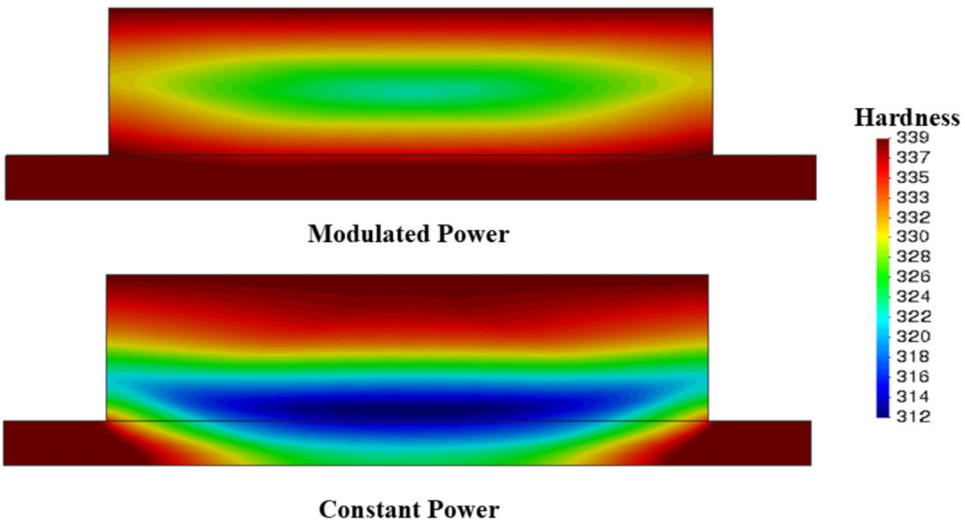


Fig. 24 Single-wall: simulation microhardness assessment

inter-layer temperature remains within a the targeted range of 500 [°C] before proceeding with subsequent deposition.

Both power strategies substantially lower the overall power input compared to constant power settings. However, the inter-layer cooling strategy maintains higher power values, as it periodically pauses the deposition every five layers to ensure effective cooling and prevent heat accumulation. Initially, the cooling intervals are shorter due to the strong heat sink effect of the substrate. As the build height increases and the influence of the substrate diminishes, the dwell intervals stabilize at approximately 60–70 s.

4.3.2 Dimensional accuracy

Dimensional accuracy was evaluated by comparing the geometry of the double-wall deposits. Figure 29 illustrates the geometric fidelity for each configuration. In Fig. 29a, b, corresponding to the modulated power setups (with and

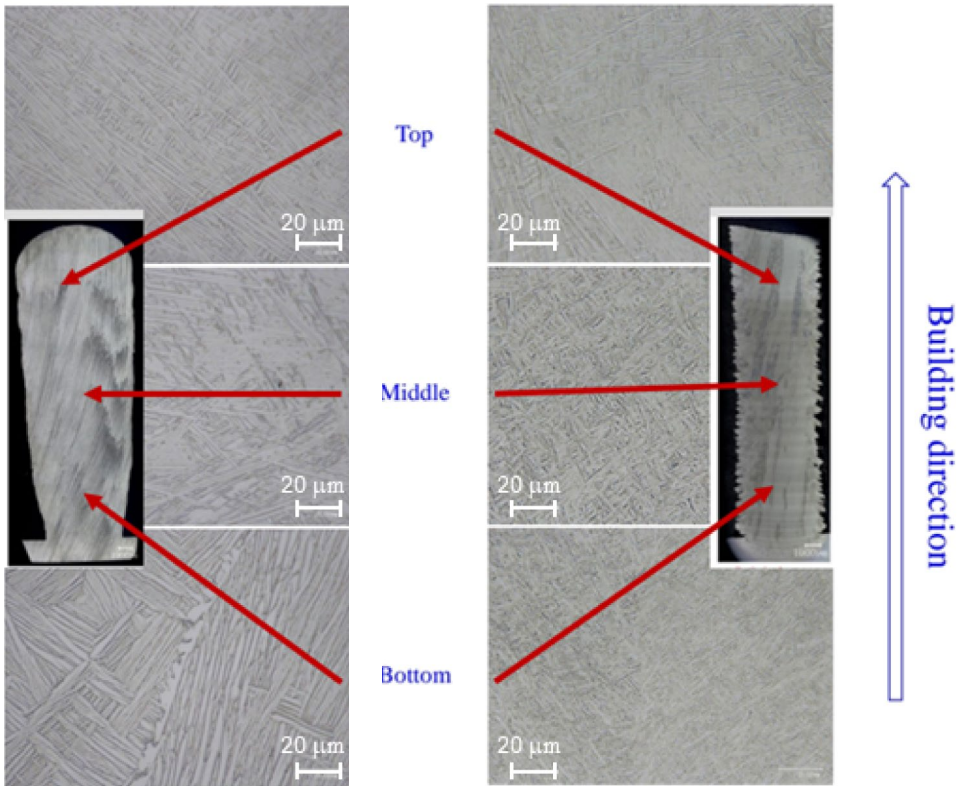


Fig. 25 Single-wall: Experimental microstructure assessment. **a** Constant power. **b** Modulated power

without inter-layer cooling), the final print of both have reached the intended height of $50\,[\mathrm{mm}]$ and exhibit relatively uniform layer buildup. Although there are minor surface texture differences, neither exhibit pronounced distortions, indicating that both modulated power strategies effectively regulated heat input and mitigated excessive thermal stresses during deposition.

Conversely, the constant power approach, illustrated in Fig. 29c, failed prematurely due to bulging at both ends of the wall. This required the early termination to prevent the nozzle from colliding with the build due to excessive height, clearly demonstrating that the unregulated high heat input led to significant issues. Consequently, the geometry remained incomplete, further underscoring the importance of power modulation for achieving reliable builds and consistent final geometries in DED processes.

As an additional remark, it is well-established that higher powder feed rates can increase surface roughness due to incomplete melting of the excess powder [73]. Therefore, power modulation strategies must consider simultaneous adjustments to the powder feed rate. Failing to reduce the powder feed rate when decreasing laser power can result in additional unmelted powder adhering to the hot walls, compromising surface quality. In this study, such adjustments were not implemented due to a lack of integrated control for synchronizing powder feed rate with dynamic power modulation in the printer system adopted.

4.3.3 Microstructure analysis

In addition to the geometrical evaluation, the microstructure of the double-wall deposits was analyzed to assess homogeneity across layers. Microstructural gradients, particularly at interfaces between differing microstructures, can induce localized plastic strain, especially when significant variations in grain size occurs [74].

Comparative microhardness measurements from the simulation (Fig. 30) and the experimental grain-size evaluations (Fig. 31) are presented for all three configurations. To capture the spatial variations in grain morphology induced by different thermal histories, measurement positions were carefully selected along the vertical extent of the build. In each specimen, samples were taken at three key elevations: $5\,[\mathrm{mm}]$ above the base, at mid-height, and $5\,[\mathrm{mm}]$ below the top of the double-wall. In this way, critical regions where temperature gradients and cooling rates vary are effectively represented.

The simulated microhardness indicates that higher values correspond to finer grains, while lower hardness values are associated with coarser grains. For the constant power

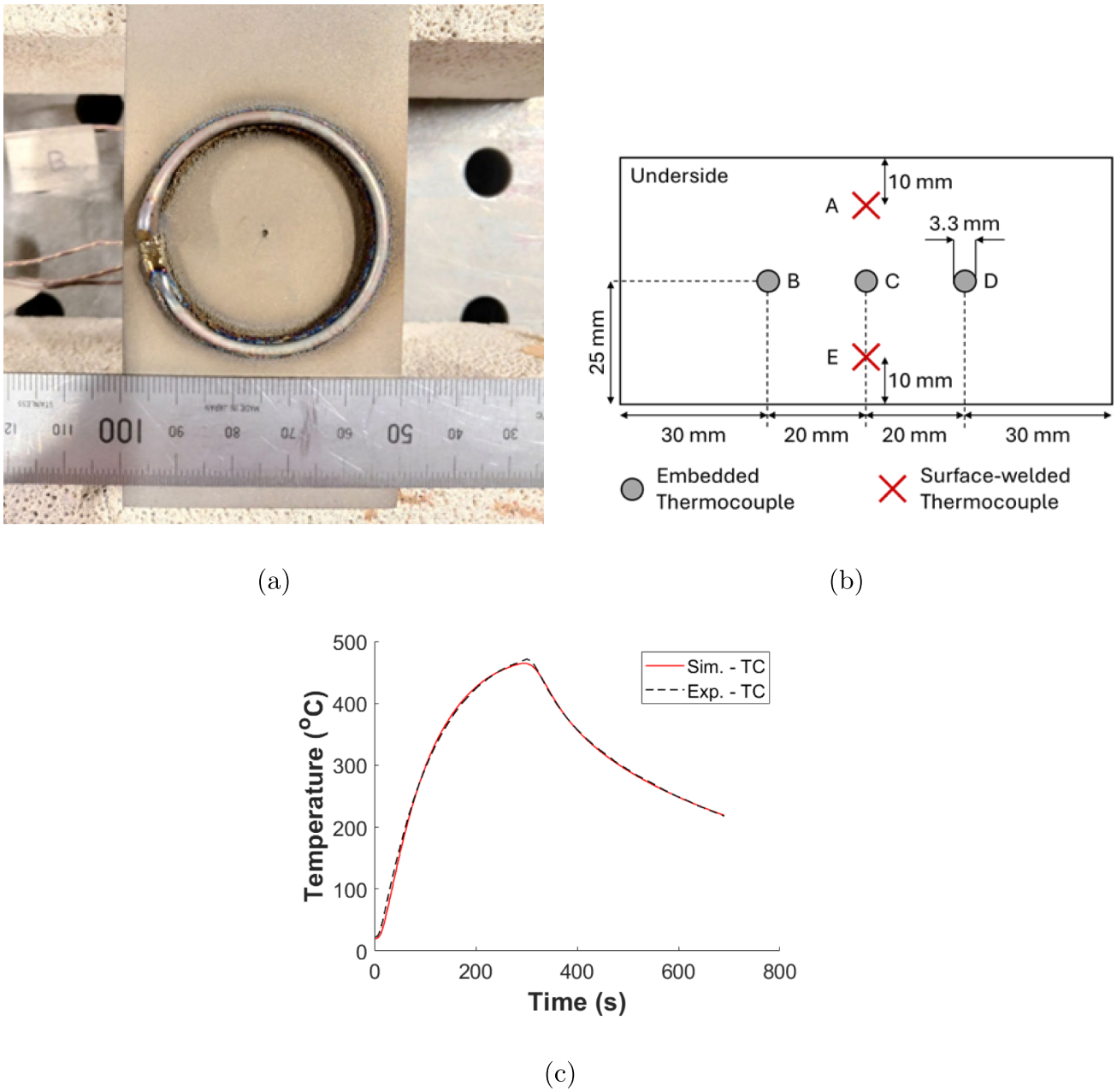


Fig. 26 Double-wall: **a** Substrate geometry and thermocouple position for the calibration setup. **b** Cylinder used for calibration supported by two refractory blocks. **c** Calibrated temperature evolution curve from thermocouple C (TC)

Table 3 Multi-track calibrated temperature-dependent HTC values

Temperature (°C)	HTC ($W/m^2\,°C$)
20	4.21
600	27.3
1200	61.64
2500	134.23

configuration, the coarsest grain region is observed at the bottom of the wall; in contrast, the modulated power configuration shifts the coarsest region toward quarter-height, and the combined modulated power with dwell time control further enhances homogeneity.

Figure 31 illustrates the experimental grain-size evaluations, revealing interesting insights into microstructural homogeneity across different vertical positions. Notably, the trends observed in the simulation microhardness results (Fig. 30) are in accordance with the experimental grain-size data, reinforcing the validity of the impact of the thermal

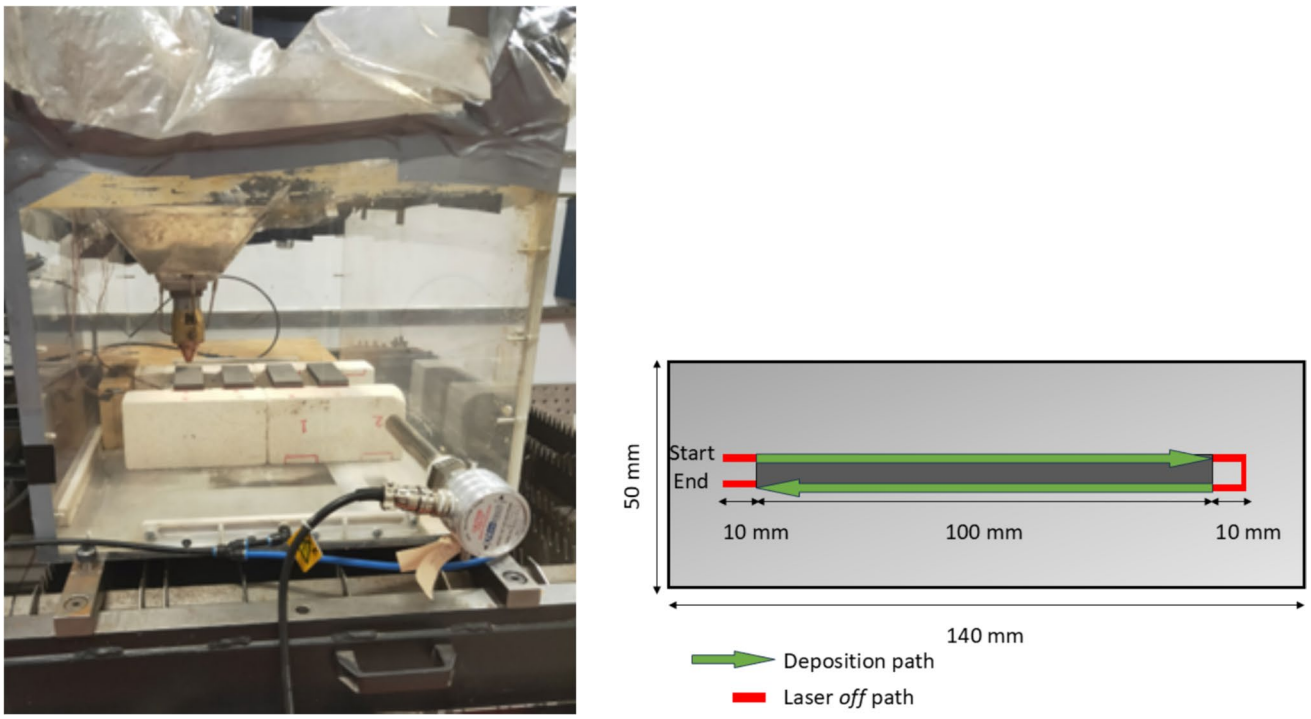


Fig. 27 Double-wall: **a** Experimental setup of the argon chamber. **b** Scanning sequence, double-wall and substrate geometries

history on microstructure. Among the evaluated strategies, the modulated power build with inter-layer cooling produced the most homogeneous grain structure across the build height, while the modulated power build without inter-layer cooling also maintained a relatively uniform microstructure, however with slightly less consistency. When comparing the available data, the modulated power builds, both with and without inter-layer cooling, provide an opportunity to evaluate the effectiveness of these strategies in maintaining uniform microstructural characteristics across the specimen, whereas the incomplete constant power build highlights potential challenges in ensuring consistency throughout the build height, as suggested by both the experimental and simulation results.

An important aspect, not directly observable in the experimental results due to the absence of substrate analysis, but clearly captured in the numerical simulations, is the ability of both power control strategies to preserve the microhardness of the substrate by minimizing thermal exposure. This effect is particularly significant in remanufacturing or repair applications, where maintaining the original properties of the underlying component is essential to ensure structural integrity and performance. This example highlights the important role of numerical simulations in providing a comprehensive understanding of the process-induced effects, offering insights that may not be readily accessible through experimental observation alone.

4.4 Remanufacturing example

To demonstrate the applicability and robustness of the proposed modeling framework in a real-world context, this final example focuses on the remanufacturing of a rotor component via the deposition of an extended axle. This case study investigates a cylindrical double-wall deposition applied to the top surface of a rotor, targeting its inner and middle radial regions. The deposition consists of 10 layers of $1.56\,[\mathrm{mm}]$ each, and the geometric complexity and thermal sensitivity of the component provide a representative industrial scenario for evaluating the performance of the framework. The rotor mesh and deposition path were constructed using the methodology described in Sect. 2.2, ensuring fidelity to the original component geometry, as illustrated in Fig. 32a, b. For both the thermal-response and mechanical-response analyses, the temperature evolution and displacements were monitored through a virtual thermocouple and a virtual displacement control point, respectively, placed at the top surface of the existing rotor. The control point, shown in magenta in Fig. 32c, enables consistent evaluation of the process-induced thermal and mechanical behavior throughout the build.

This analysis explores the effectiveness of power modulation strategies in controlling melt-pool behavior, and evaluates the resultant thermal profiles, mechanical responses and microstructural evolution. Through this, the case study emphasizes the critical role of the simulation tool in

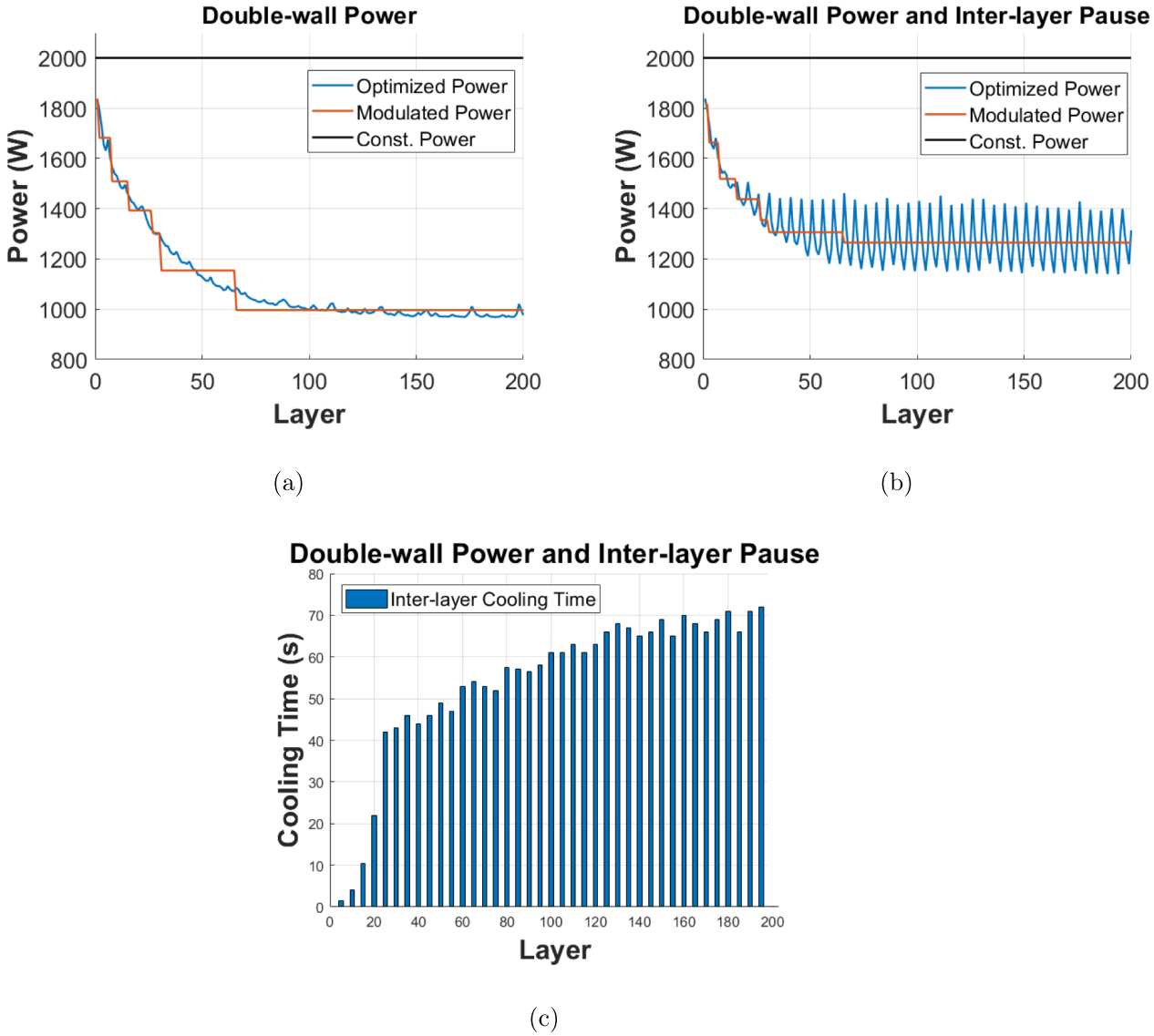


Fig. 28 Double-wall: Power profiles and inter-layer cooling strategy. **a** Modulated power profile without inter-layer cooling; **b** modulated power profile with inter-layer cooling; **c** corresponding cooling durations applied every 5 layers in the inter-layer cooling configuration

optimizing additive remanufacturing parameters and ensuring component qualification.

4.4.1 Power modulation assessment

In this example, the deposition process employs a modulated power strategy to construct the axle extension on the top surface of the rotor. The power profile was generated using the methodology described in Sect. 3.1, targeting an average melt-pool penetration depth of approximately half that observed under constant power conditions. This approach aims to minimize excessive melt-pool penetration and ensure controlled energy input throughout the build height. The modulation strategy adopts a per-layer approach, whereby the power remains constant during each individual layer but varies from one layer to the next. This results in a stepped power profile along the build direction, as shown in Fig. 33. Compared to in-plane deposits, the curved surface of the rotor presents additional challenges in maintaining a consistent thermal input, due to localized differences in heat dissipation and surface curvature. Consequently, power modulation plays a crucial role in avoiding excessive melting or under-deposition. The optimized profile helps stabilize the melt-pool and promotes a uniform distribution of heat input across the cylindrical geometry.

The modulated power profile in Fig. 33 begins with a pronounced reduction over the first three layers, reflecting a strong heat-sink effect from the substrate. At the fourth layer, power briefly increases to compensate for the loss of substrate cooling as the build height grows. Thereafter, the

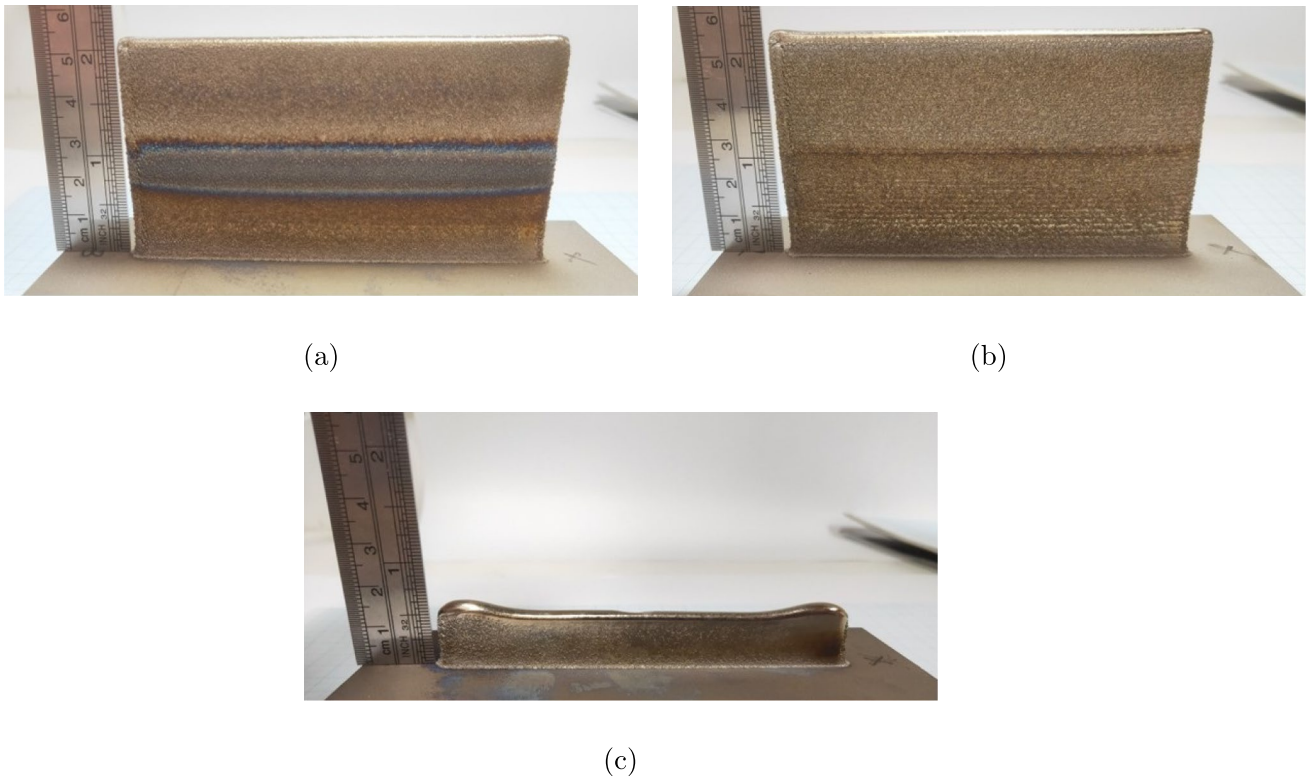


Fig. 29 Double-wall: Final geometries. **a** Modulated power profile without inter-layer cooling. **b** Modulated power profile with inter-layer cooling. **c** Constant power (failed prematurely)

power profile gradually decreases in a stepped, per-layer fashion, to prevent the growth of the melt-pool penetration depth.

4.4.2 Thermal-response analysis

The virtual thermocouple positioned as depicted in Fig. 32c records the thermal history for the constant and modulated power setups. Figure 34 presents (a) the temperature evolution and (b) the corresponding cooling rate for both the constant-power and modulated-power strategies. Under constant power, the peak temperature rises rapidly with each layer-and fluctuates by over $100\,[^{\circ}C]$ between successive layers-while the cooling rate (b) exhibits pronounced spikes exceeding $10\,[^{\circ}C/s]$. By contrast, the modulated-power approach limits peak temperatures by 20-30 $[^{\circ}C]$ and attenuates inter-layer temperature oscillations. The resulting cooling-rate profile is both lower in magnitude and markedly smoother throughout the build, with peak rates reduced up to $40\,\%$ compared to the constant-power case.

4.4.3 Mechanical-response analysis

Mechanical boundary conditions are applied to the red-shaded region in Fig. 35, fully constraining displacements ($u_x = u_y = y_z = 0$). Displacement evolution at the virtual control point, presented in Fig. 32c, is compared in Fig. 36-radial in (a) and vertical in (b). Under constant power, radial displacement peaks at approximately $0.15\,[\mathrm{mm}]$, whereas the modulated-power strategy limits this peak to around $0.11\,[\mathrm{mm}]$. Similarly, vertical displacement under constant power reaches around $0.14\,[\mathrm{mm}]$, compared to $0.11\,[\mathrm{mm}]$ with modulation. After deposition, both strategies converge toward a stable offset of 0.03–0.04 $[\mathrm{mm}]$, but the reduced peak excursions under modulated conditions indicate enhanced geometric fidelity and lower process-induced distortion.

Displacement-contour plots in Fig. 37 further illustrate these differences. Subfigures (a) and (b) show the modulated-power case at the end of deposition and after cooling, respectively; (c) and (d) present the analogous views for constant power. The modulated configuration produces a more uniform displacement field, particularly during deposition, with peak displacements limited to $0.31\,[\mathrm{mm}]$. In contrast, the constant-power case exhibits greater distortion, reaching peak displacements of approximately $0.35\,[\mathrm{mm}]$.

Figure 38 presents residual stress distributions with subfigures (a) modulated and (b) constant power supplies. Both strategies develop tensile peak stresses (up to $500\,[\mathrm{MPa}]$) at the interface between the original rotor and the newly deposited axle extension.

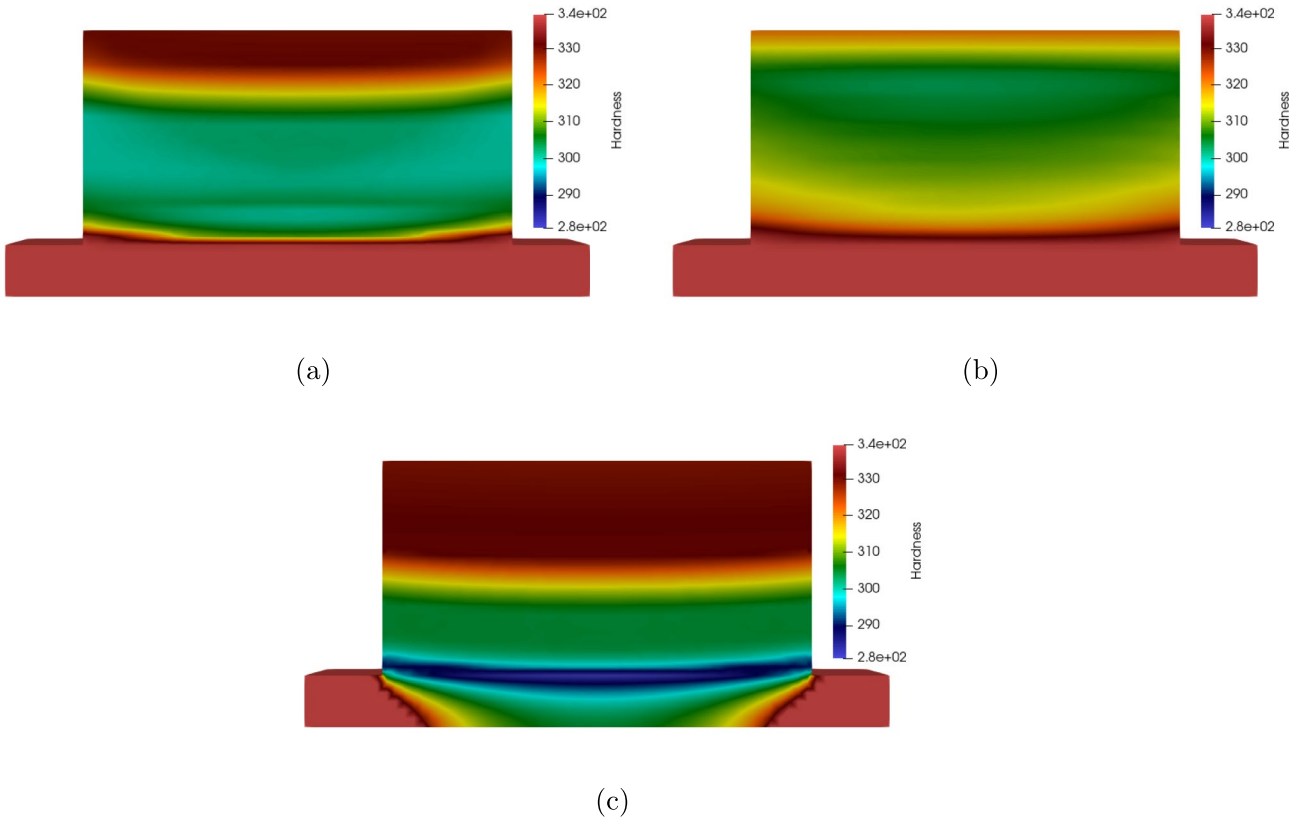


Fig. 30 Double-wall: numerical results for the microhardness. **a** Modulated power profile without inter-layer cooling. **b** Modulated power profile with inter-layer cooling. **c** Constant power

The modulated power supply, by enforcing tighter temperature control and minimizing substrate heat accumulation, produces steeper thermal gradients at each layer interface. As illustrated in Fig. 37a, the controlled thermal regime reduces warpage during deposition compared to the constant power supply in Fig. 37c. However, the cooler substrate temperature increases the stiffness of the underlying structure, which introduces greater residual stresses, particularly between layers and at the rotor-deposit interface. Despite an overall lower thermal input, the modulated strategy induces higher residual stresses in both the newly deposited axle extension and the pre-existing rotor body, as depicted in the stress contours of Fig. 38.

4.4.4 Microstructure analysis

The microstructural evolution of the deposited axle extension was quantified in terms of the fractions of α_{M}, α_{W}, β , and the Vickers microhardness. Figure 39 presents these distributions for the modulated-power configuration in subfigures (a)–(d) and for the constant-power configuration in subfigures (e)–(h).

Under modulated power the α_{M} fraction is higher at the rotor-deposit interface and decreases smoothly toward the top of the deposition. Elevated α_{M} denotes extensive martensitic transformation, which increases strength and hardness at the expense of ductility. The α_{W} fraction likewise peaks at the substrate boundary-indicating a pronounced Widmanstätten plate morphology that enhances toughness-before tapering toward the top layers. In contrast, the β fraction is minimal (0-1 %) at the interface, rises to a peak of 6 % mid-height, and settles around 4 % near the top; retained β phase improves ductility while reducing overall hardness. Finally, the microhardness values shows a narrow distribution about the mean, confirming a consistently uniform mechanical performance throughout the axle extension.

Contrarily, the constant power supply, depicted in Fig. 39e–h, displays a broader, asymmetric distribution of the α_{M} fraction with its peak at the rotor-deposit interface is reduced compared to the modulated case, and it fluctuates irregularly through the mid-height layers, indicating uneven martensitic transformation. The α_{W} fraction is substantially higher in the upper layers compared to the modulated power supply, reflecting extensive Widmanstätten growth driven by sustained thermal exposure. Conversely, the β fraction remains elevated (peaking near the substrate-deposit interface) and sustains higher β fraction toward the top of the deposition, signifying pockets of retained ductile phase

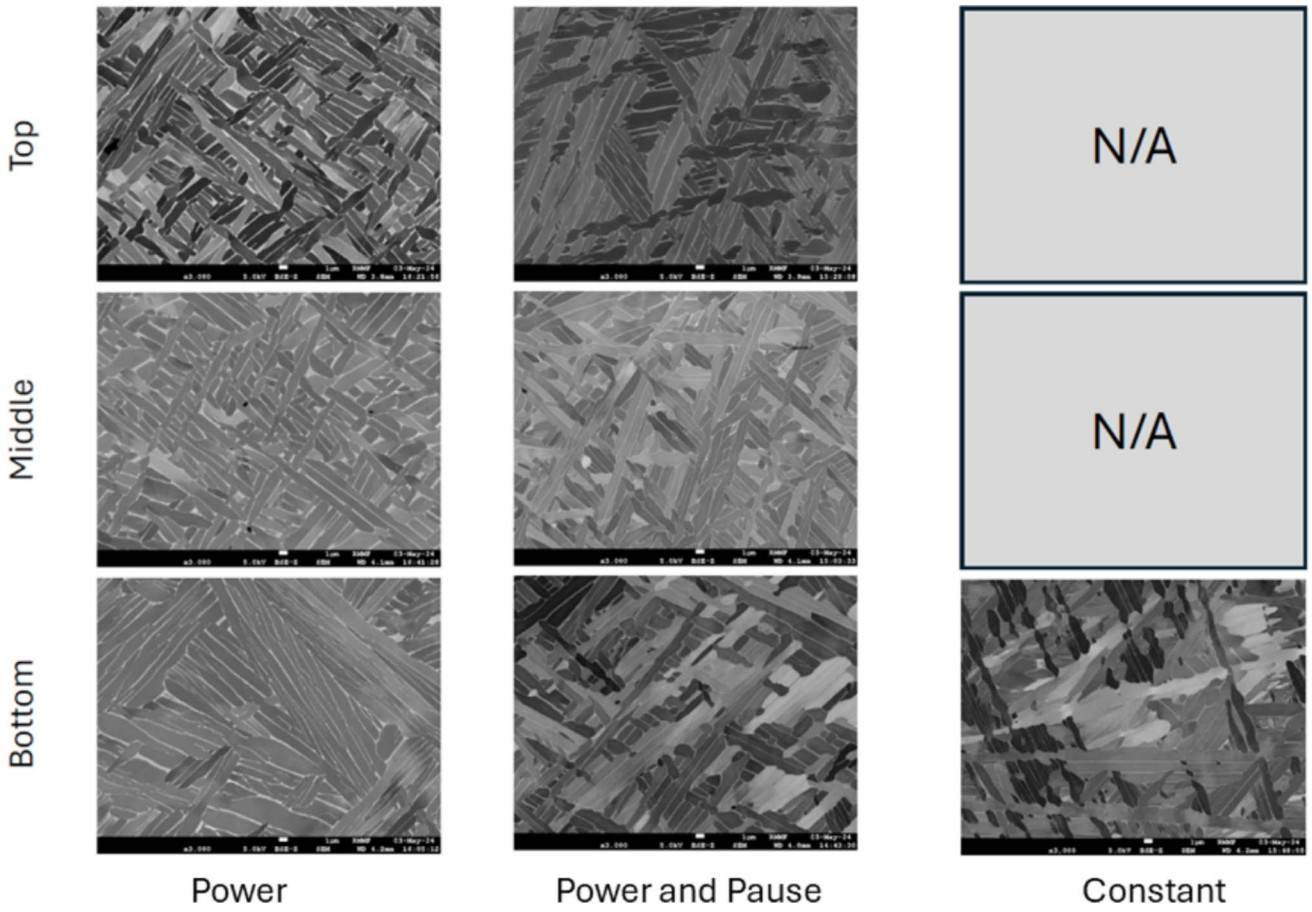


Fig. 31 Double-wall: experimental microstructure magnified ×3000 for the three settings

amid incomplete transformation. Finally, microhardness values exhibit a significantly wider spread, mirroring the pronounced phase heterogeneity.

Overall, the modulated-power strategy yields a markedly more uniform microstructure-both in phase composition and hardness-compared to constant power. By contrast, the constant-power approach produces significant variability in phase fractions and hardness, underscoring its detrimental effect on microstructural consistency and mechanical reliability. This enhanced homogeneity under modulated power is essential for ensuring predictable fatigue resistance and long-term integrity in remanufactured components.

5 Conclusions

This study investigates the qualification of repair and remanufacturing operations using DED for Ti-6Al-4V components through high-fidelity simulations supported by experimental validation. The strong agreement between simulated and measured data confirms the accuracy of the thermal model and the effectiveness of the developed numerical framework and control strategies. Process qualification is achieved through dynamic power modulation and inter-layer dwell time, while component qualification is assessed based on dimensional accuracy, residual stresses, distortions, and microstructural evolution. The work provides a comprehensive evaluation of power control strategies in DED, highlighting essential trade-offs between power control and final part quality across different performance metrics.

- Process and component qualification are analyzed separately, showing that improvements in one do not necessarily translate to the other. Both require distinct optimization strategies depending on the target outcome-buildability, microstructural uniformity, dimensional accuracy, or residual stress reduction.
- Power modulation enhances dimensional accuracy by stabilizing melt-pool behavior and mitigating heat accumulation, resulting in consistent wall geometry and reduced distortion, especially in thin substrates and complex geometries.
- Adaptive power control promotes smoother thermal profiles with lower peak temperatures and fewer cooling-rate fluctuations, minimizing overheating and oxidation risks. However, it can induce steeper thermal gradients

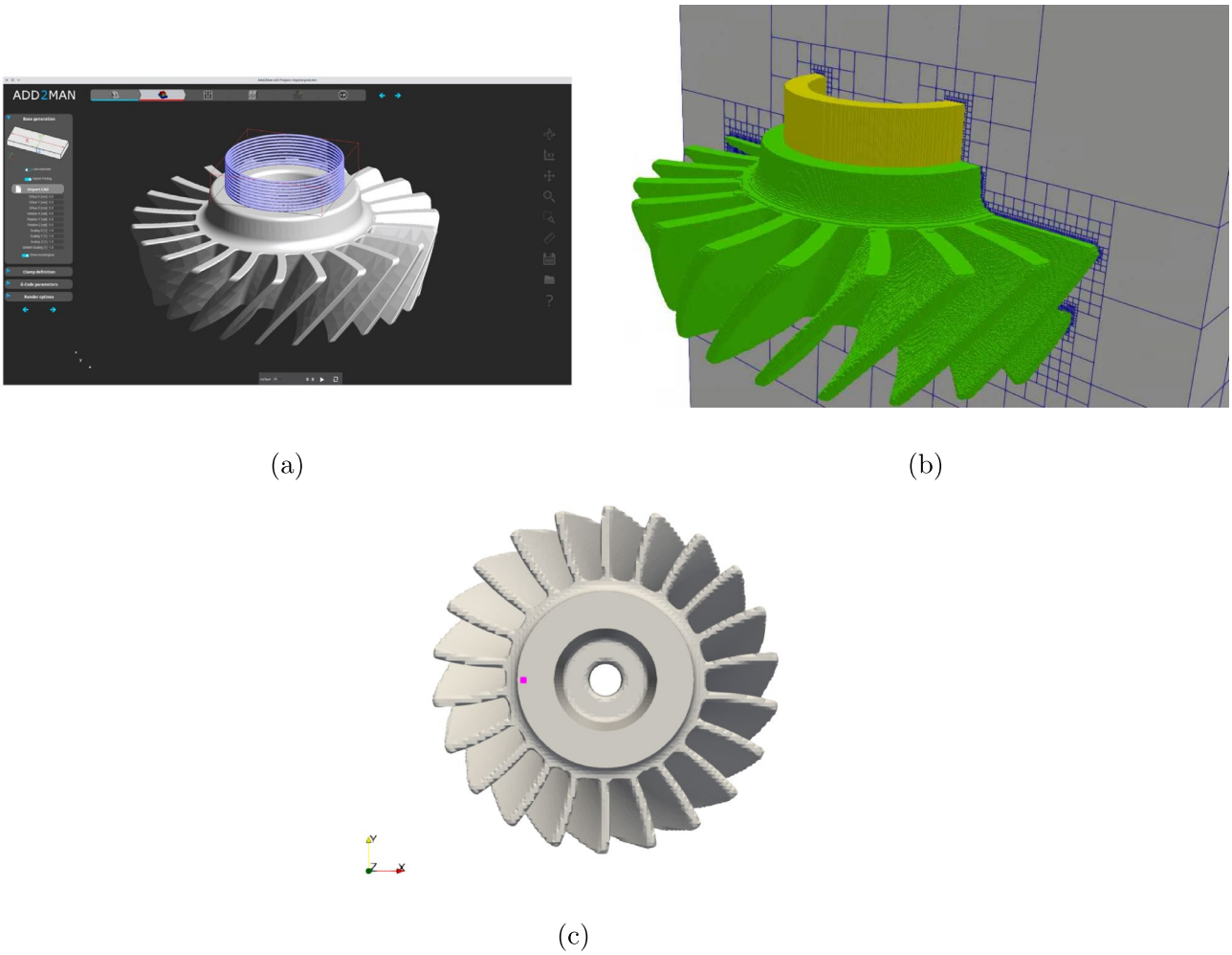


Fig. 32 Remanufacturing: AM built extension for a rotor. **a** Add2Man CAD visualization. **b** Mesh. **c** Virtual thermocouple and displacement control point

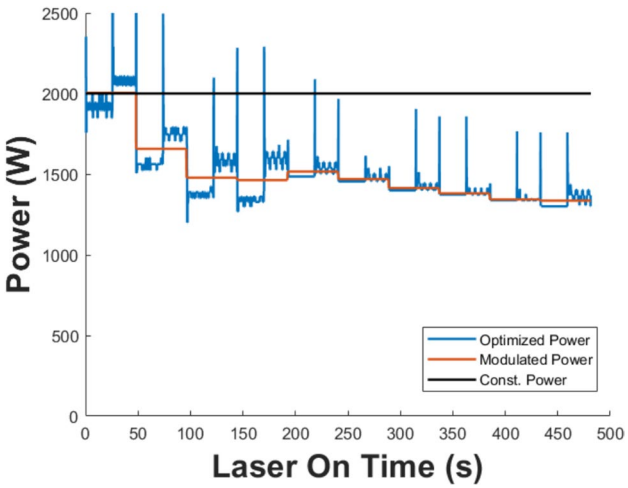


Fig. 33 Remanufacturing: Modulated power profile curve

in low-inertia substrates, potentially increasing local stress concentrations.

- The mechanical response indicates that modulated power slightly reduces global deformation but may increase residual stresses near the substrate interface due to localized thermal gradients.
- Power modulation promotes uniform phase distribution and hardness, yielding fine martensitic structures with limited retained β phase. While inter-layer dwell time further enhances microstructural uniformity, it also increases inter-layer stresses due to deposition on a colder surface.
- In contrast, constant-power deposition leads to coarser microstructures, greater phase segregation, and higher hardness variability, compromising overall mechanical integrity.
- Overall, dynamic power control significantly improves thermal regulation, dimensional stability, and microstructural consistency, establishing it as an effective,

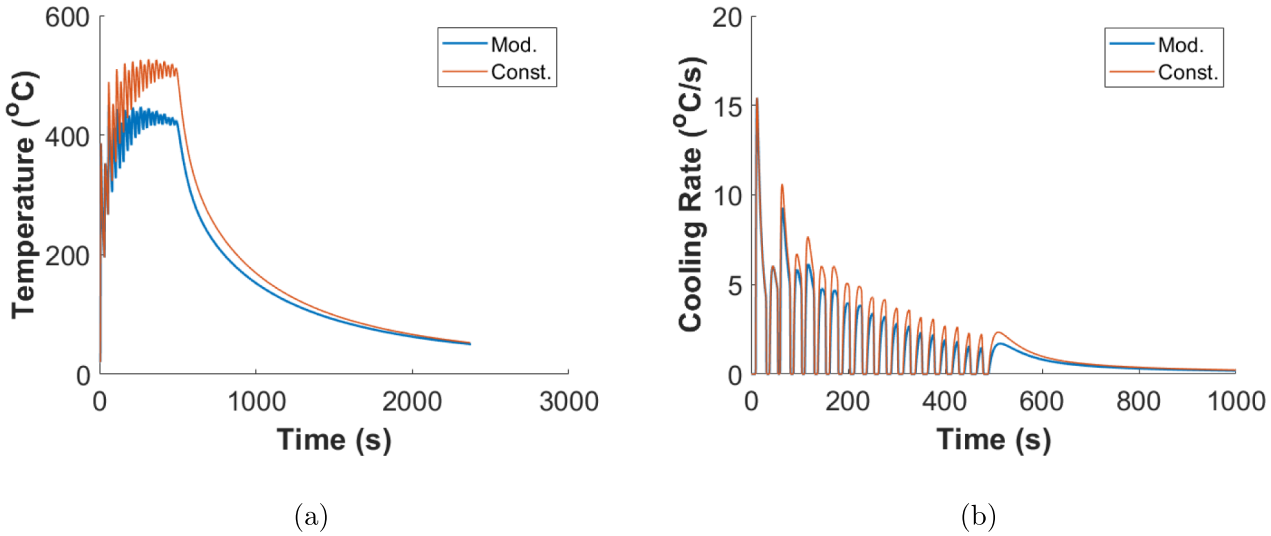


Fig. 34 Remanufacturing: Comparison between modulated and constant power setups at the virtual thermocouple coordinate. **a** Temperature evolution. **b** Cooling rate

Fig. 35 Remanufacturing: Mechanical boundary conditions applied to red area. **a** Cross section. **b** Bottom view



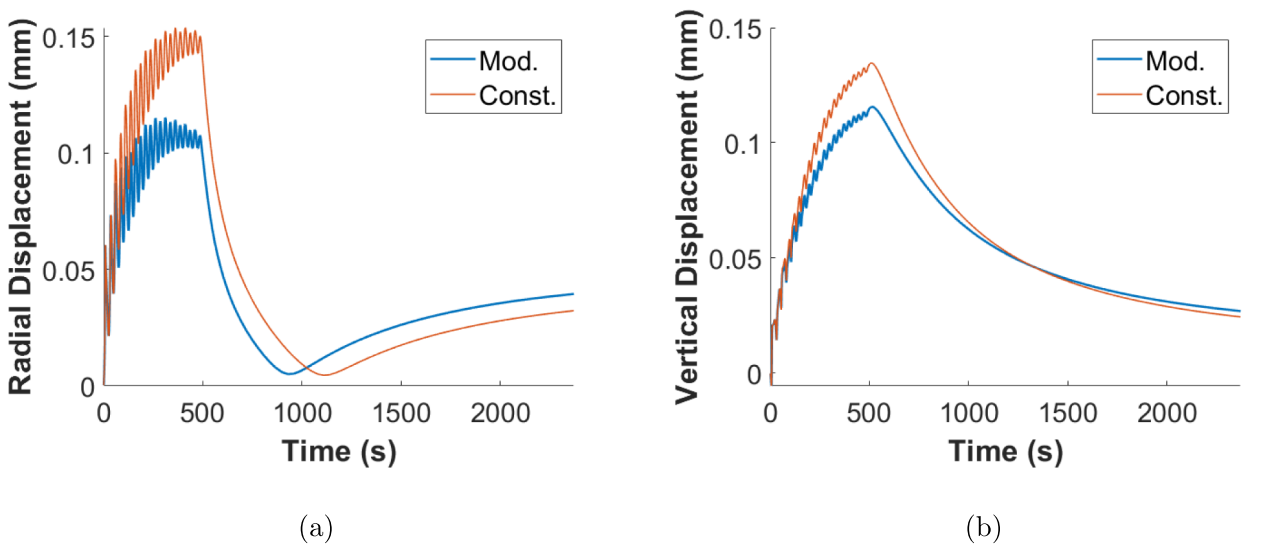


Fig. 36 Remanufacturing: Displacement evolution comparison between the constant and modulated power. **a** Radial displacement. **b** Vertical displacement

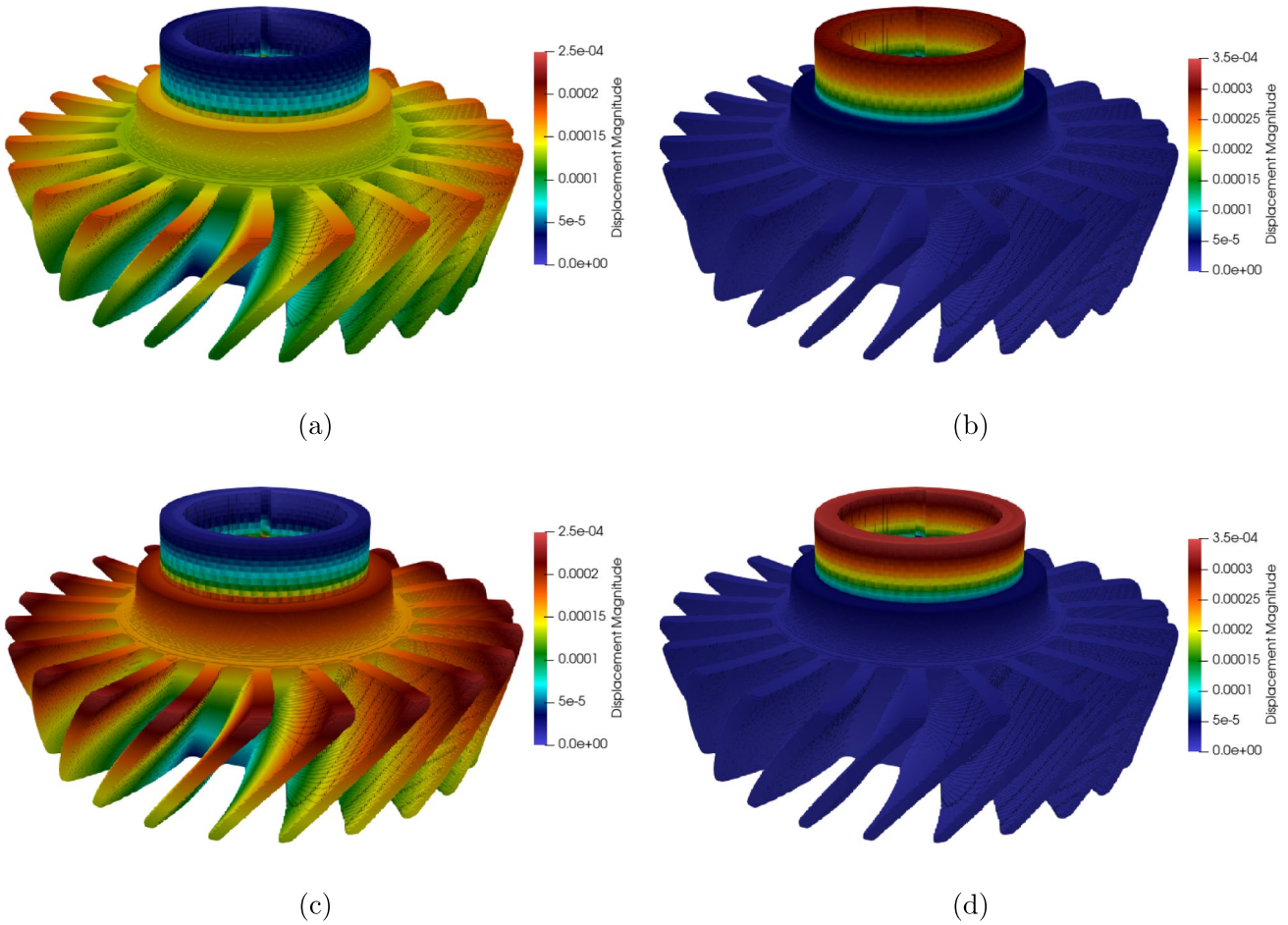


Fig. 37 Remanufacturing: Displacement contour plots. **a** Final deposition time-step and **b** post-cooling stage for the modulated power setup; **c** final deposition time-step and **d** post-cooling stage for the constant power setup

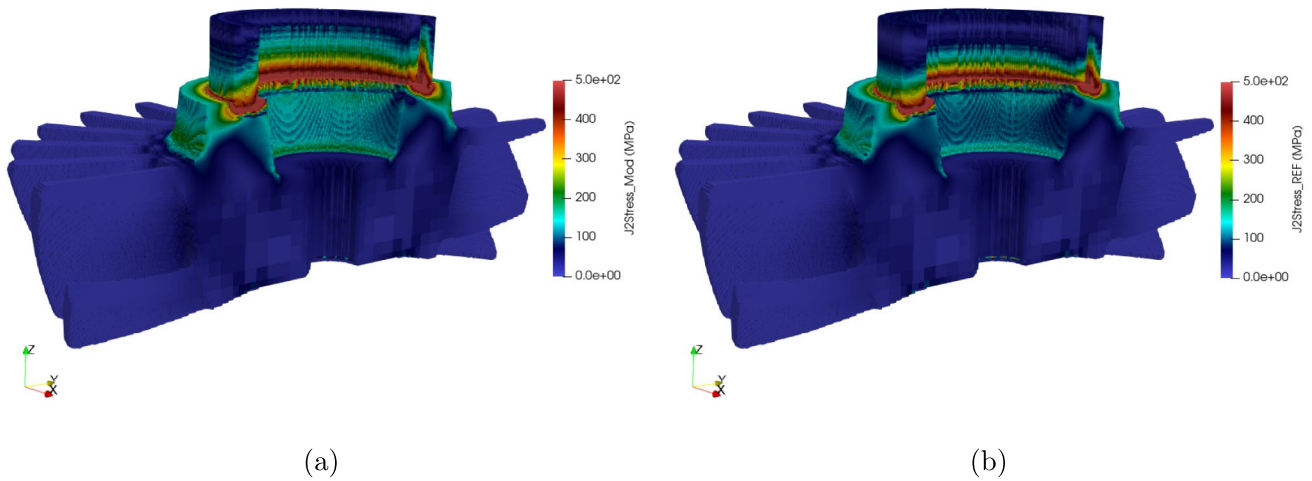


Fig. 38 Remanufacturing: Residual stresses on a section cut of the rotor after the post-cooling stage. **a** Modulated power and **b** constant power supplies

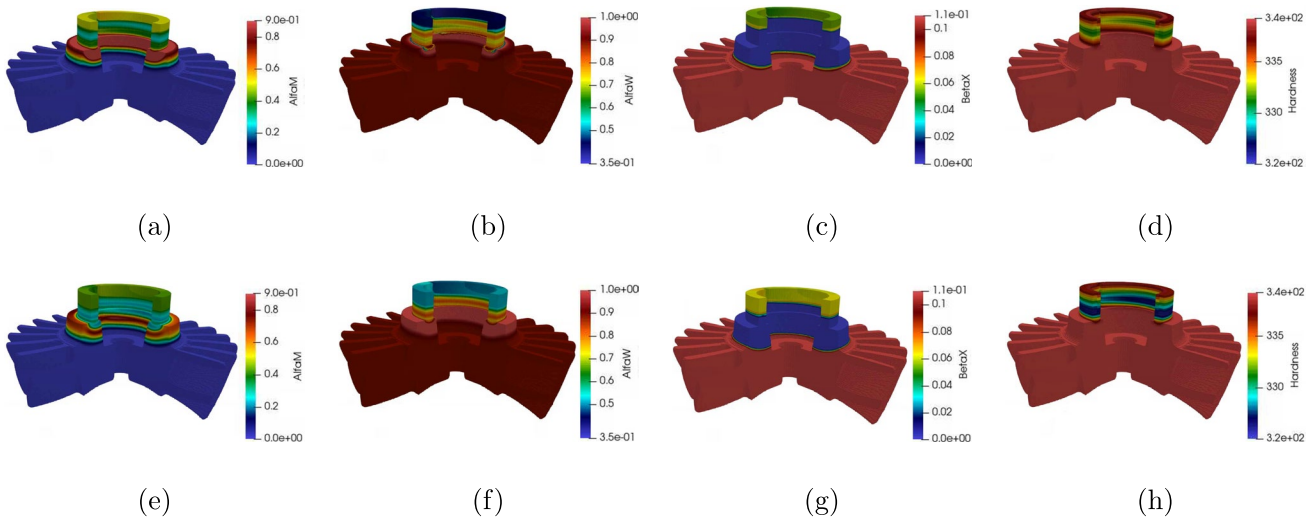


Fig. 39 Remanufacturing: Subfigures **a**–**d** show the microstructural parameters under modulated power conditions-specifically, the fractions of α_M, α_W, β and the measured microhardness, respectively. Subfigures **e**–**h** display the corresponding results under constant power mode

simulation-driven approach for the qualification of DED-based repair and remanufacturing processes.

- Melt-pool penetration is readily accessible numerically but difficult to measure in situ. For experimental validation, melt-pool area or morphology metrics obtained from thermal imaging provide a practical alternative.

Future research focuses on investigating the influence of scanning sequence strategies on both process and component qualification to further optimize thermal management, microstructural evolution, and mechanical performance in DED. In addition, the extension of the proposed framework to large-scale components will be investigated by restricting the refinement to the vicinity of the deposition path and heat-affected zone, while using progressively coarser non-conforming equivalent meshes in regions far from the repair area to reduce computational cost without affecting accuracy in the region of interest.

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Declarations

Conflict of interest The authors declare no Conflict of interest.

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